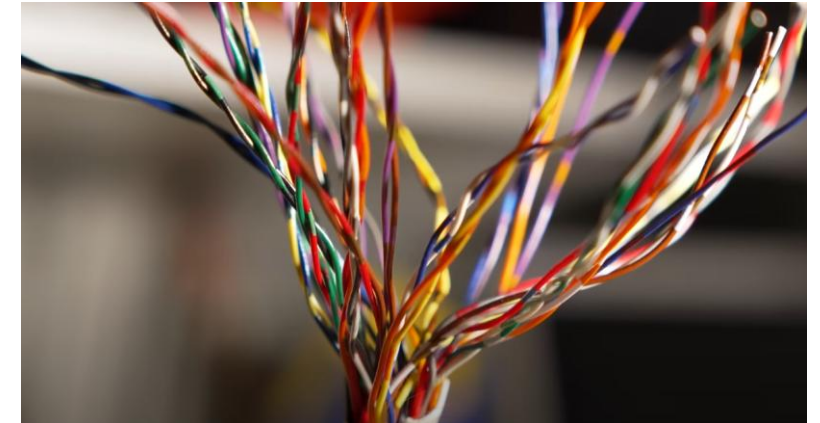
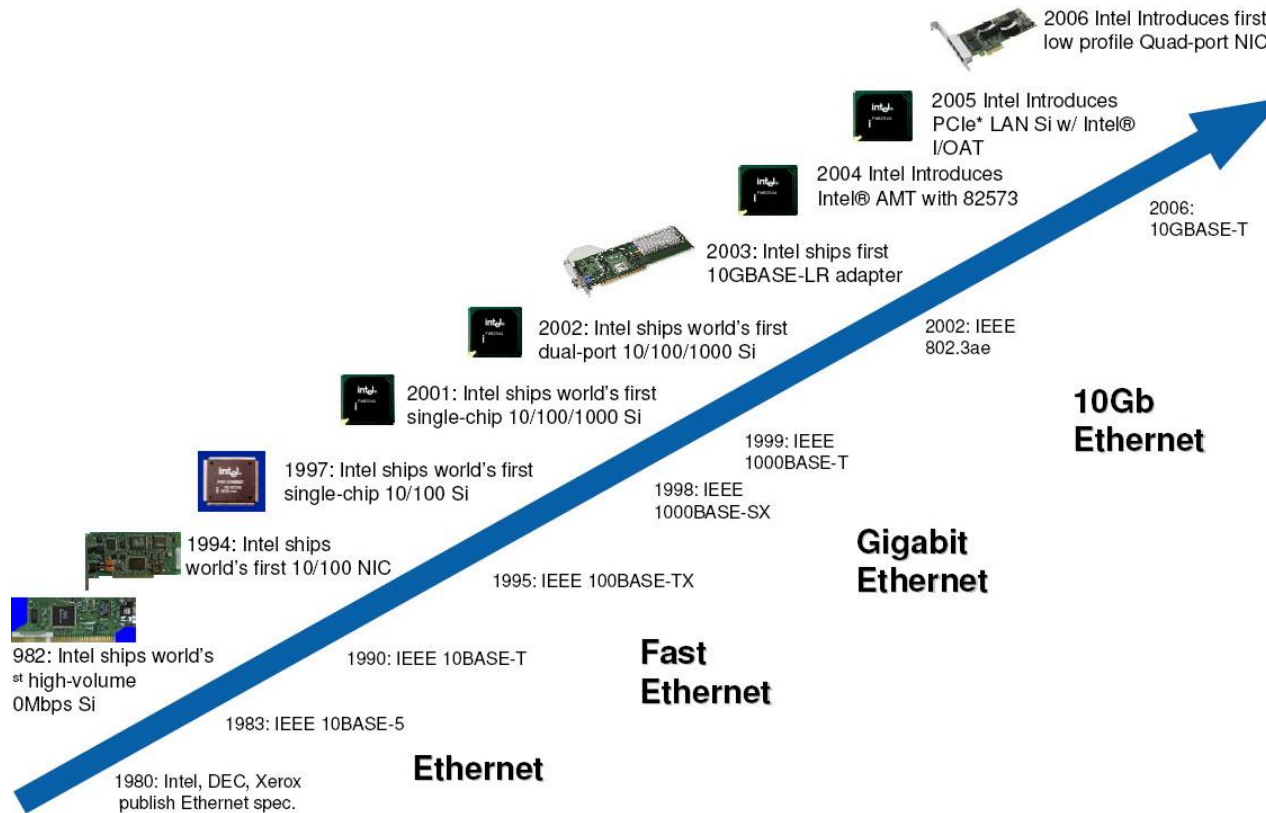


Network Fundamentals

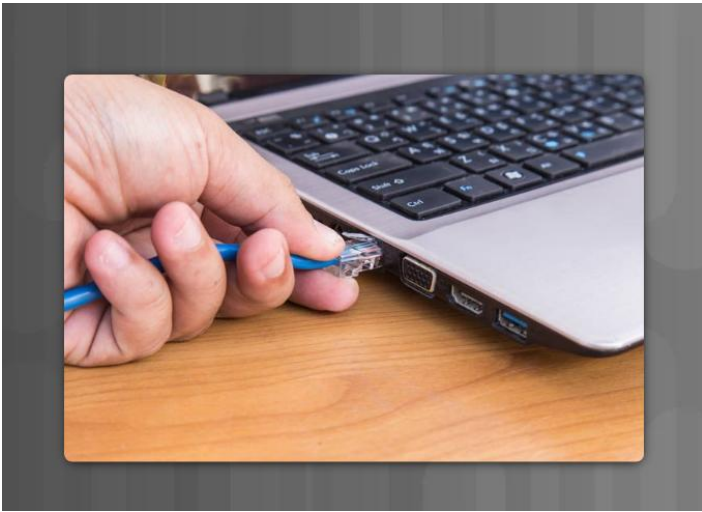
Network access and the physical layer

History of Ethernet



<https://www.youtube.com/watch?v=f8PP5IHsL8Y>

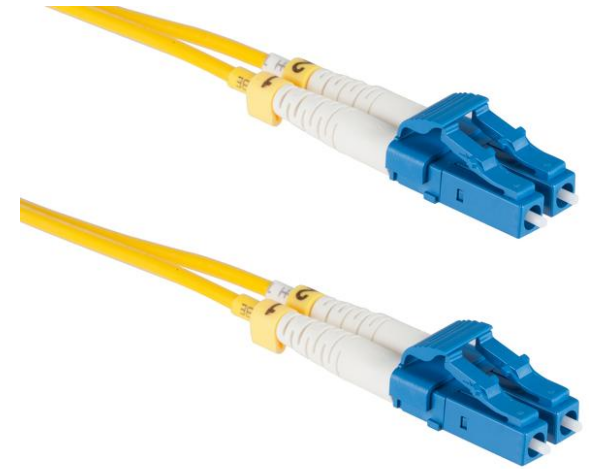
Types of connections



Wired connection



Wireless connection

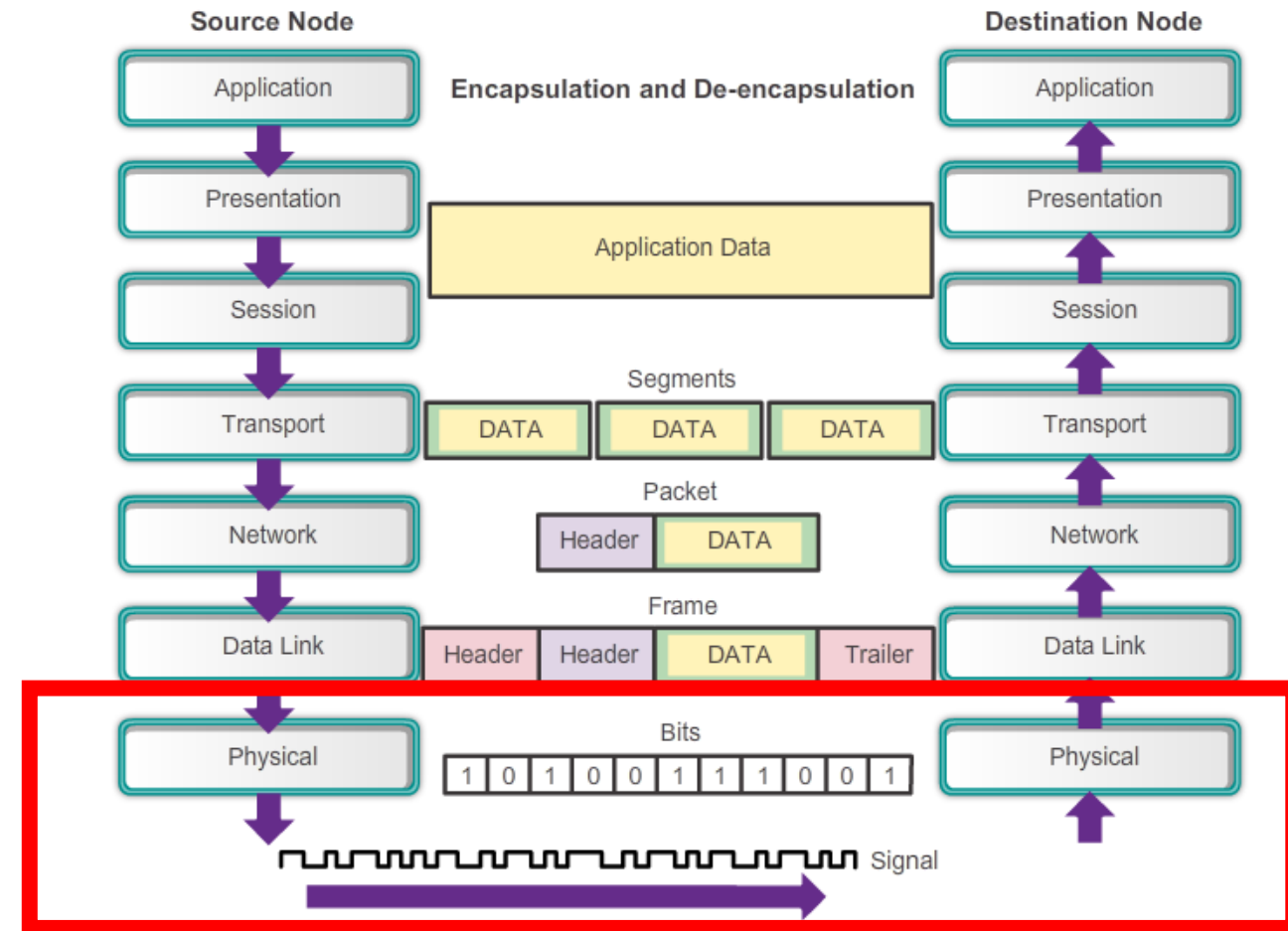
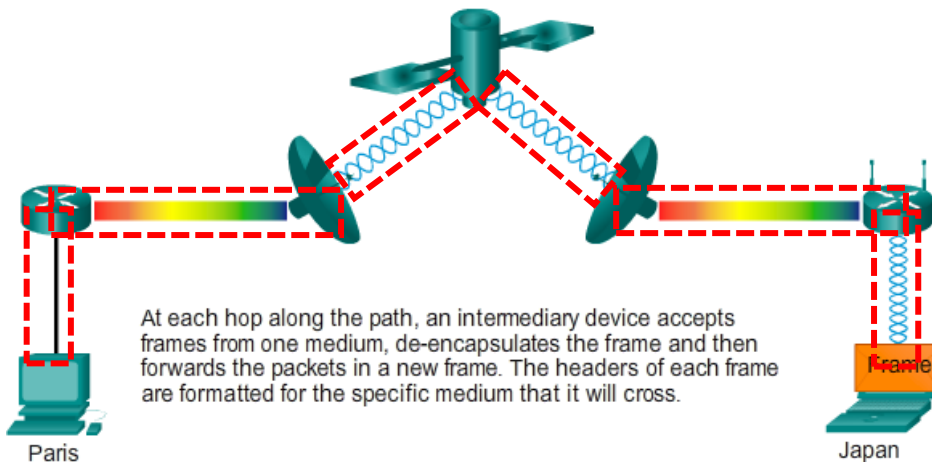


Fiber

The Physical Layer

How will a stream of bits be transmitted over the medium

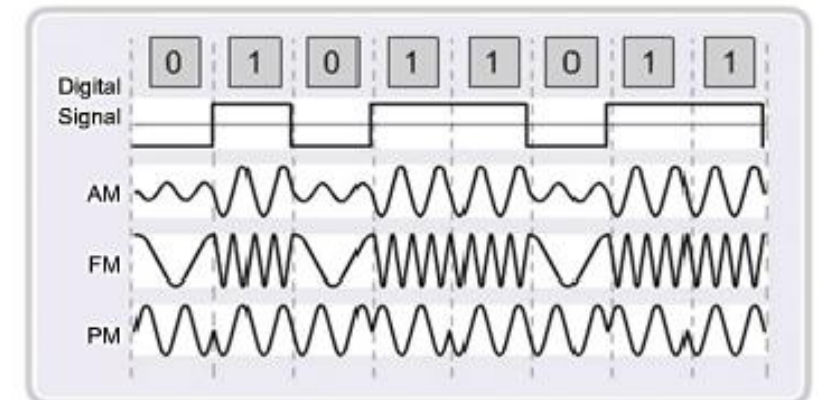
Describes the point-to-point transmission of bits



Physical Layer Media

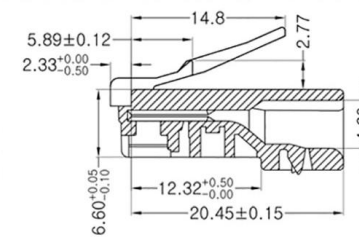
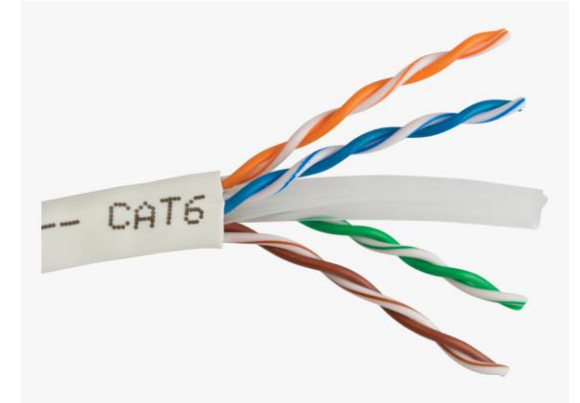
There are three basic forms of network media.

- **Copper cable:** The signals are patterns of electrical pulses.
- **Fiber-optic cable:** The signals are patterns of light.
- **Wireless:** The signals are patterns of microwave transmissions.



Physical layer components

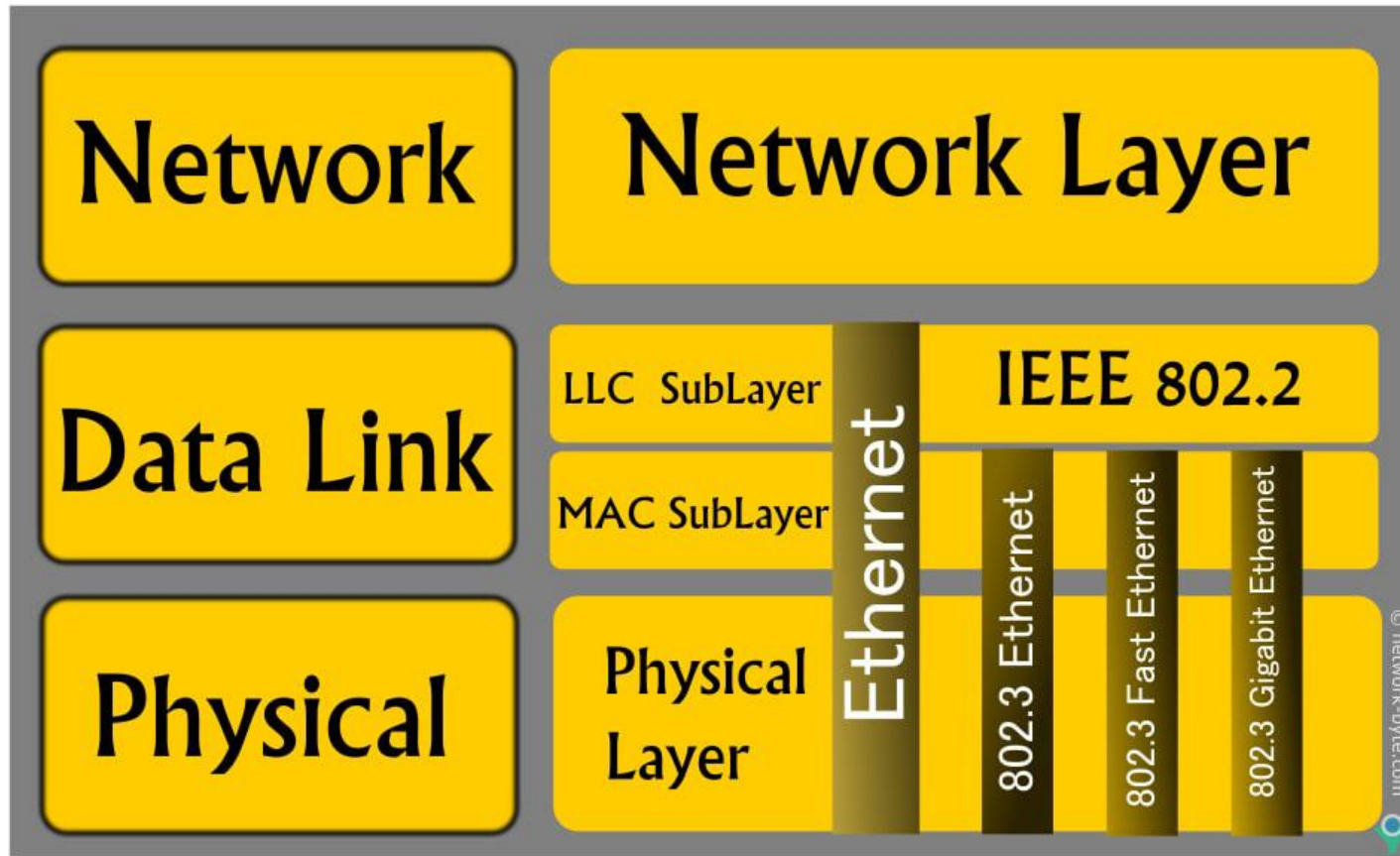
- Physical and electrical properties of the **media** (cables)
- Mechanical properties of the **connectors** (interface)
- **Encoding** of the bits
- **Data Rate** (bits per second) and **Synchronization**
- **Topologies** (Mesh, Star, Ring, Bus)
- **Transmission** mode (Simplex, Half Duplex, Full Duplex)
- **Access methods** on shared medium



Physical layer protocols

- **Ethernet (IEEE 802.3):** Used in wired LANs for defining physical connections and transmission standards.
- **Wi-Fi (IEEE 802.11):** Defines wireless communication standards.
- **Bluetooth:** Facilitates short-range wireless communication.
- **DSL (Digital Subscriber Line):** Manages high-speed internet access over telephone lines.
- **USB (Universal Serial Bus):** Defines physical interfaces for device-to-device communication.

Ethernet



List of ethernet protocols

OSI	Ethernet		
Data Link Layer	Logical Link Control (LLC)		
	Medium Access Control (MAC)		
Physical Layer	<u>Standard Ethernet</u>	<u>Fast Ethernet</u>	<u>Gigabit Ethernet</u>
	10Base5	100BaseTX	1000BaseT
	10Base2	100BaseT4	1000BaseLX
	10BaseT	100BaseFX	
	10BaseFX		

Types of Ethernet

Ethernet Type	Bandwidth	Cable Type	Duplex	Maximum Distance
10Base-5	10 Mbps	Thicknet Coaxial	Half	500 m
10Base-2	10 Mbps	Thinnet Coaxial	Half	185 m
100Base-TX	10 Mbps	Cat3/Cat5 UTP	Half	100 m
100Base-TX	100 Mbps	Cat5 UTP	Half	100 m
100Base-FX	200 Mbps	Cat5 UTP	Full	100 m
100Base-FX	100 Mbps	Multimode Fiber	Half	400 m
1000Base-T	200 Mbps	Multimode Fiber	Full	2 km
1000Base-TX	1 Gbps	Cat5e UTP	Full	100 m
1000Base-SX	1 Gbps	Cat6 UTP	Full	100 m
1000Base-LX	1 Gbps	Multimode Fiber	Full	550 m
10GBase-CX4	1 Gbps	Single-Mode Fiber	Full	2 km
10GBase-T	10 Gbps	Twin-axial	Full	100 m
10GBase-LX4	10 Gbps	Cat6a/Cat7 UTP	Full	100 m
10GBase-LX4	10 Gbps	Multimode Fiber	Full	300 m
10 Mbps	10 Gbps	Single-Mode Fiber	Full	10 km

Encoding

Encoding

- = a method of converting a stream of data bits into a predefined code.
- **Codes = groupings of bits** used to provide a predictable pattern that can be recognized by both the sender and the receiver.

Encoding methods include:

- **Manchester encoding:** A 0 is represented by a high to low voltage transition and a 1 is represented as a low to high voltage transition. This type of encoding is used in older versions of Ethernet, RFID and Near Field Communication.
- **Non-Return to Zero (NRZ):** This is a common means of encoding data that has two states termed “zero” and “one” and no neutral or rest position. A 0 may be represented by one voltage level on the media and a 1 might be represented by a different voltage on the media.

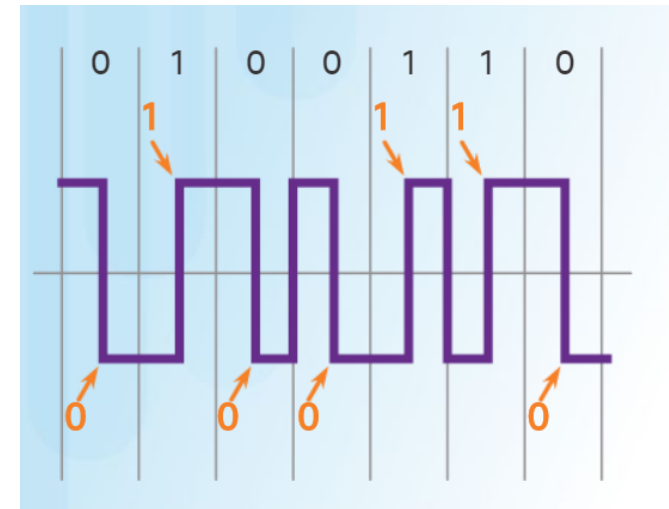
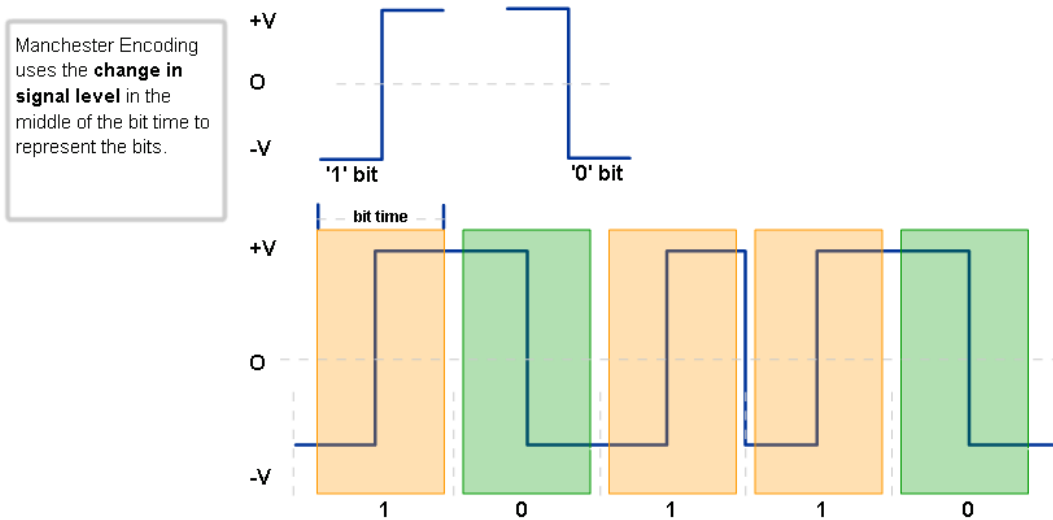
Note: Faster data rates require more complex encoding, such as 4B/5B, however, explanation of these methods is beyond the scope of this course.

Encoding

Manchester code (Ethernet 10BASE-T)

- Transition from a low voltage to a high voltage represents a bit value of **1**
- Transition from a high voltage to a low voltage represents a bit value of **0**.

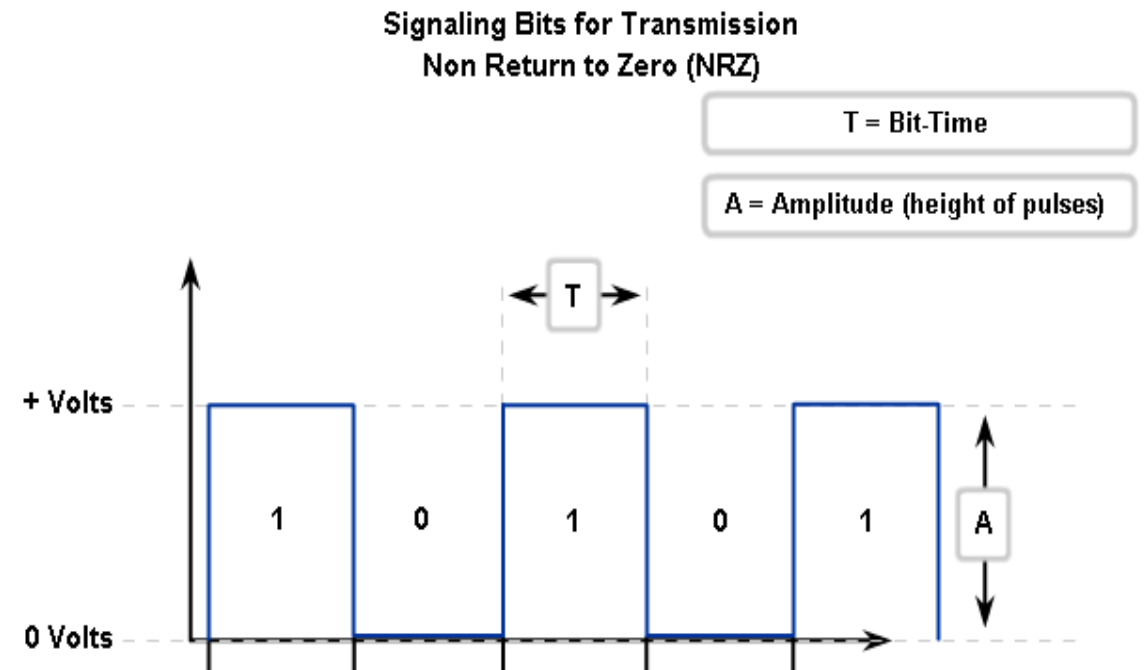
Signaling Bits for Transmission
Manchester Encoding



Encoding

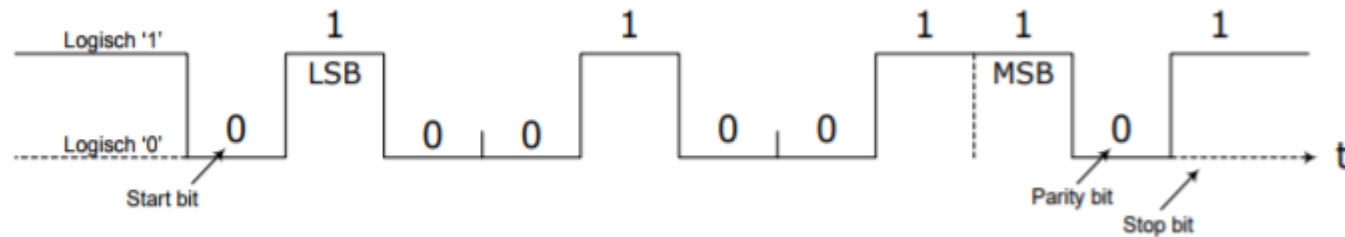
NRZ (Non-return to zero) (Ethernet 100BASE-T)

- A low voltage value represents a logical 0
- High voltage value represents a logical 1



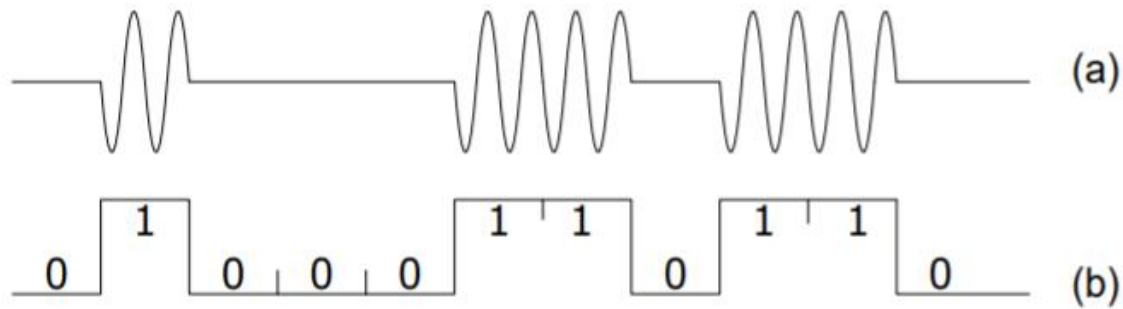
ENCODING

- 8 bit (1 byte) transmission code (+ start bit, stop bit, parity bit)
- Keep sender and receiver in sync

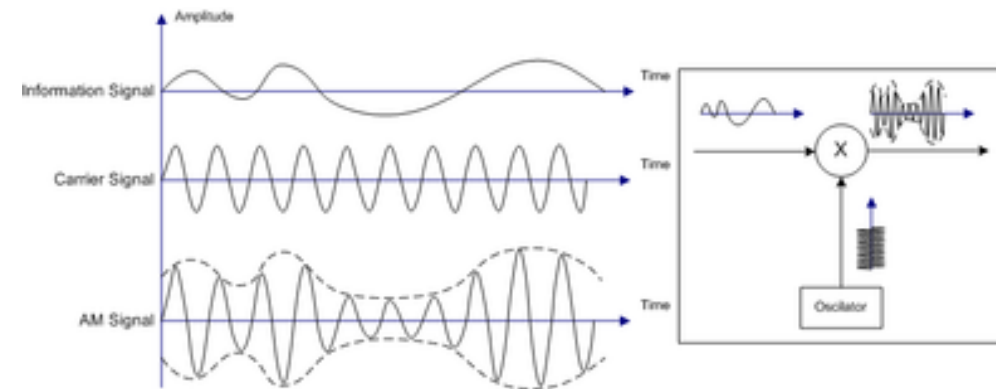


Modulation

Digital information can be modulated as an analog signal.



- Less susceptible to electrical disturbances
- Multiplex multiple signals
- Sending signals over longer distances



Modulation

Signaling

The physical layer must generate the electrical, optical, or wireless signals that represent the "1" and "0" on the media. The method of representing the bits is called the signaling method. Signals can be transmitted in one of two ways:

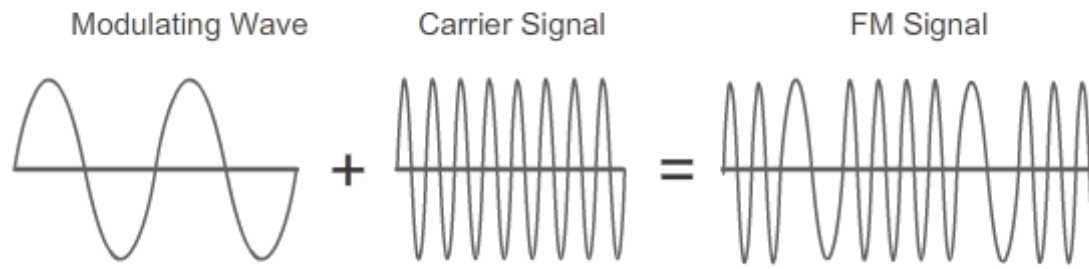
- **Asynchronous:** Data signals are transmitted without an associated clock signal. Therefore, frames require start and stop indicator flags. (RS232, RS485,...)
- **Synchronous:** Data signals are sent along with a clock signal which occurs at evenly spaced time durations referred to as the bit time. (UART, SPI, USB)

Signaling use modulation techniques :

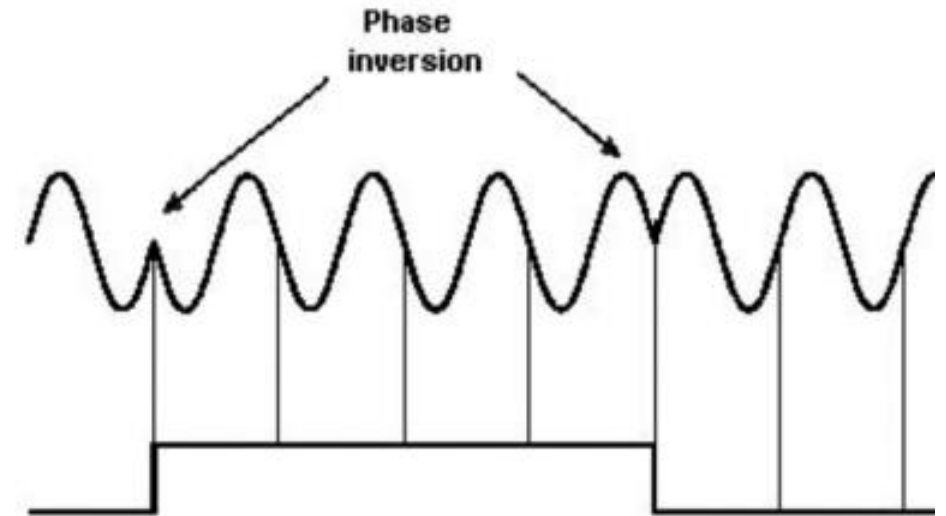
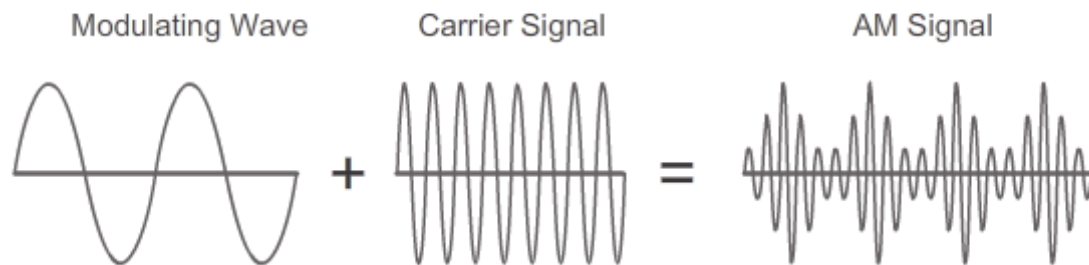
- **Frequency modulation (FM):** A method of transmission in which the carrier frequency varies in accordance with the signal.
- **Amplitude modulation (AM):** A transmission technique in which the amplitude of the carrier varies in accordance with the signal.
- **Pulse-coded modulation (PCM):** A technique in which an analog signal, such as a voice, is converted into a digital signal by sampling the signal's amplitude and expressing the different amplitudes as a binary number. The sampling rate must be at least twice the highest frequency in the signal.

Modulation

Frequency Modulation (FM)

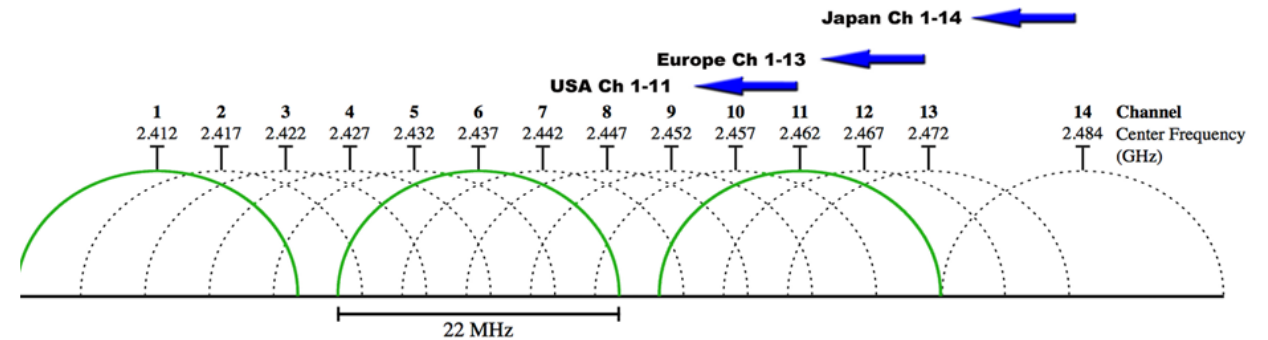
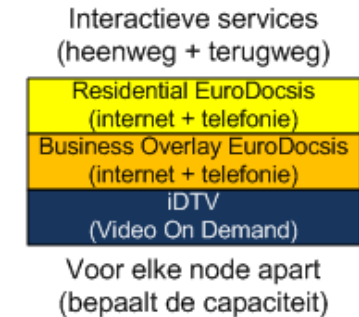
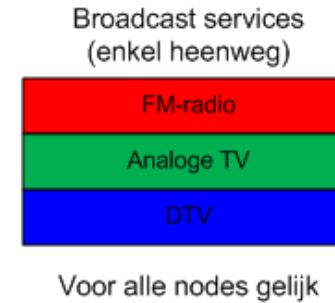
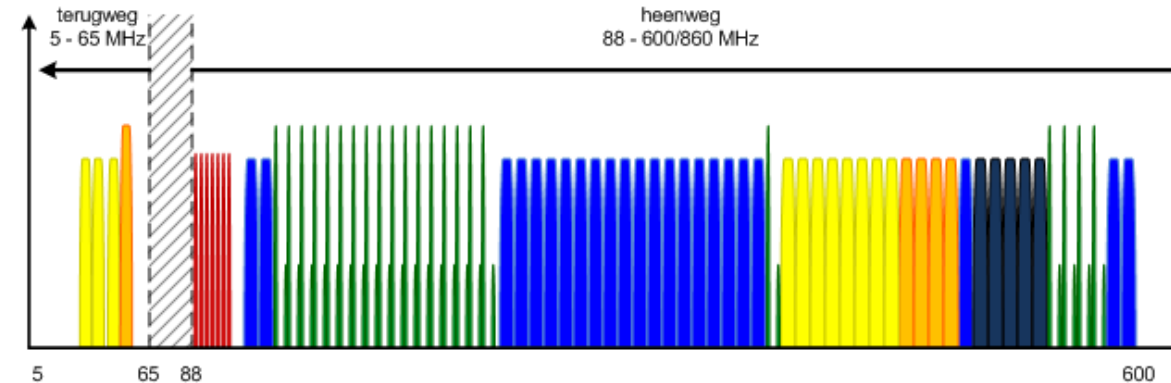


Amplitude Modulation (AM)



channels

- **Physical channel** = twisted pair cable, coax cable, EM, fiber cable
- **Logic channel** = modulated channels on physical channel



Bandwidth

The practical bandwidth of a network is determined by a combination of factors:

- The properties of the physical media
- The technologies chosen for signaling and detecting network signals

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	kbps	1 kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

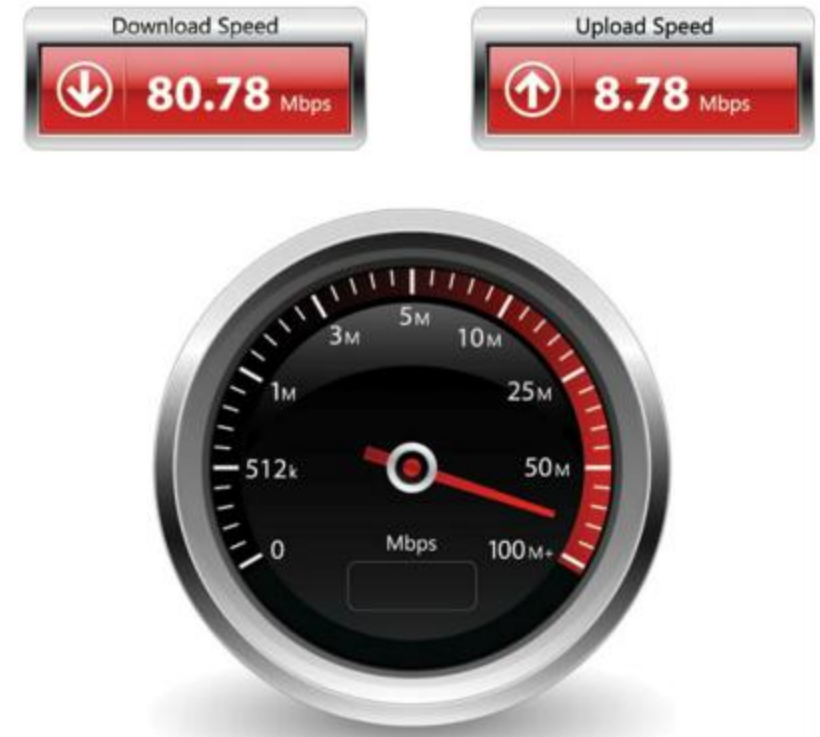
Throughput

Throughput is the measure of the transfer of bits across the media over a given period of time.

Due to a number of factors, throughput usually does not match the specified bandwidth in Physical layer implementations

In a network with multiple segments:

- throughput cannot be faster than the slowest link of the path from source to destination. Even if all or most of the segments have high bandwidth
- only one segment in the path with low bandwidth can create a bottleneck to the throughput of the entire network
- <http://speedtest.telenet.be/>



Goodput

= the measure of usable data transferred over a given period of time

= the most interesting measure to network users

→ measures the effective transfer of user data between Application layer entities, such as between a source web server process and a destination web browser device.

Goodput accounts the bits devoted to the protocol as overhead.

Goodput = throughput – traffic overhead for establishing sessions, acknowledgements, and encapsulation.

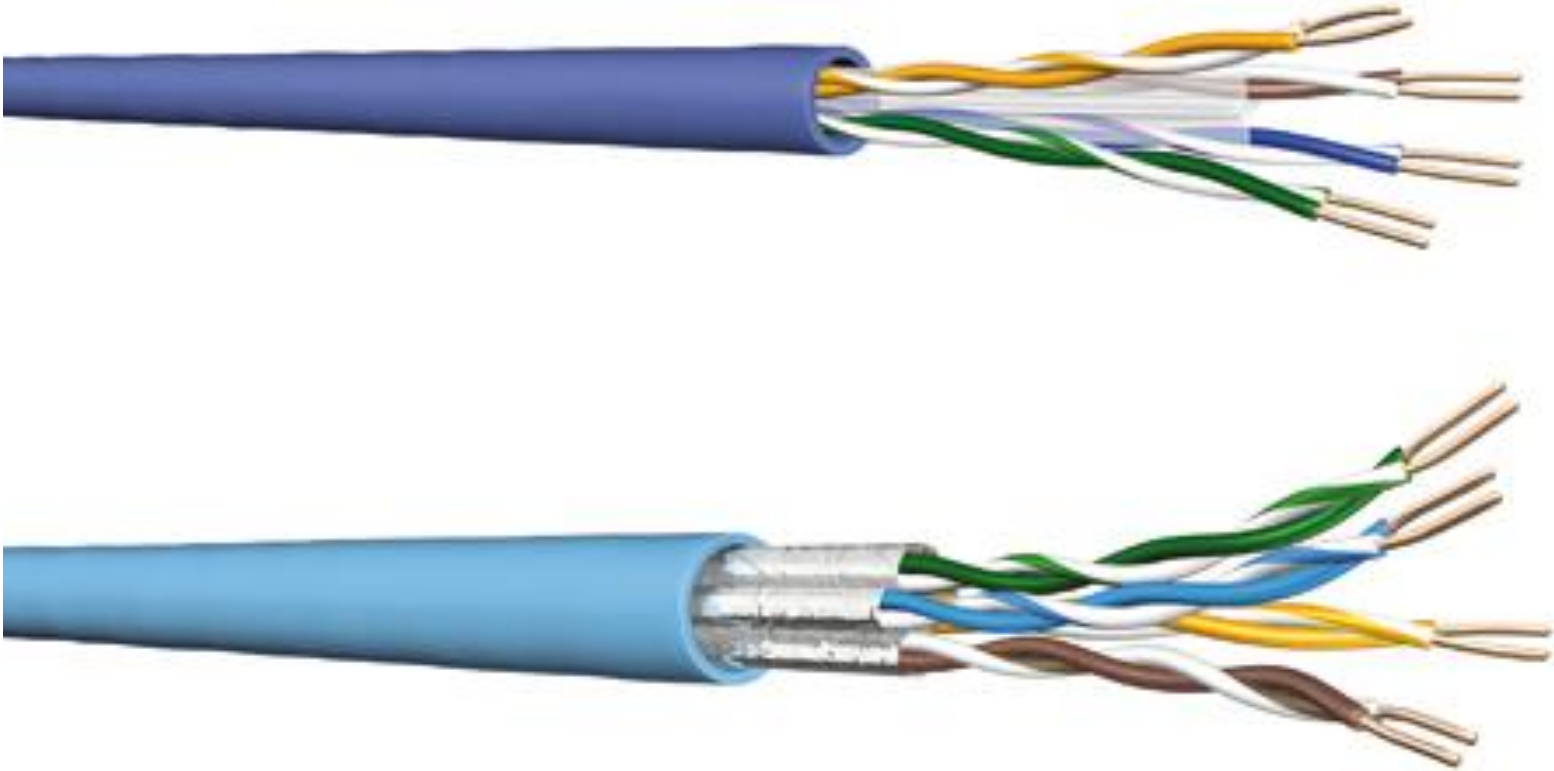
Example: two hosts on a LAN transferring a file

The bandwidth of the LAN is 100 Mbps.

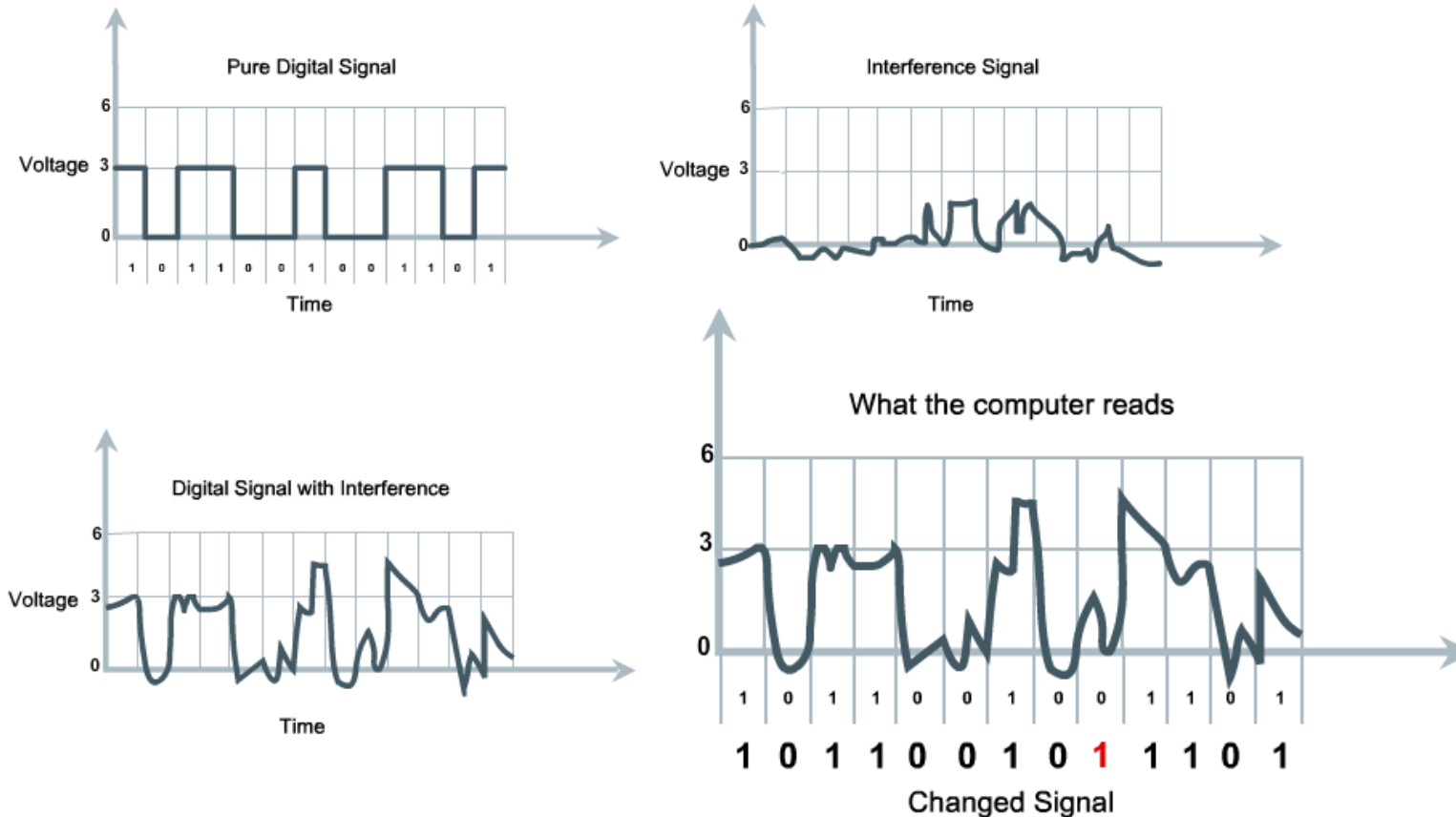
Due to the sharing and media overhead the throughput between the computers is only 60 Mbps.

With the overhead of the encapsulation process of the TCP/IP stack, the actual rate of the data received by the destination computer, goodput, is only 40Mbps.

Copper Cabling



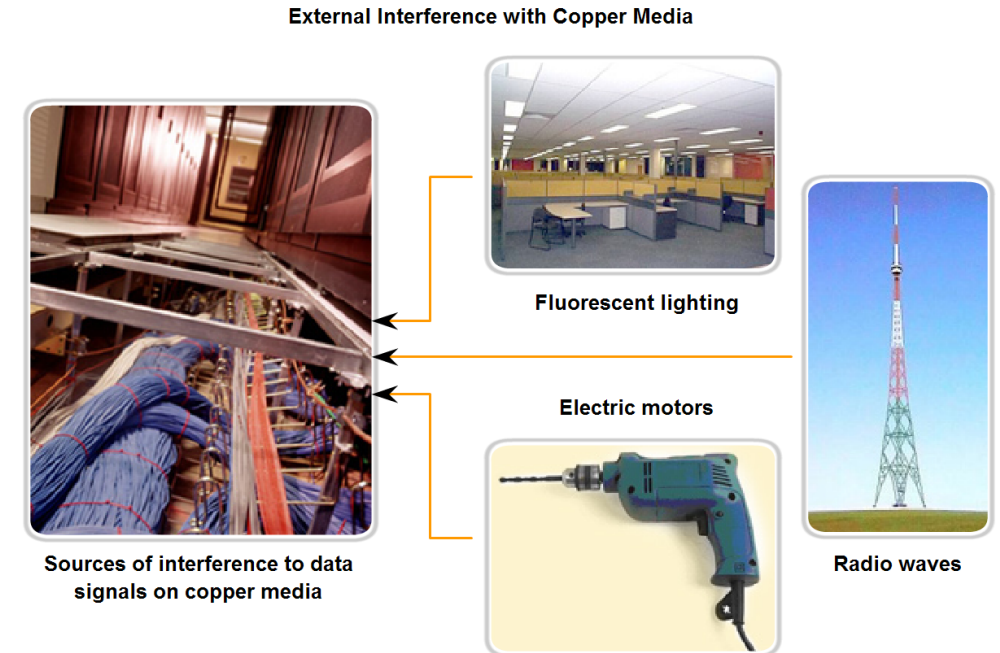
Characteristics of Copper Media



Characteristics of Copper Media

The timing and voltage values of the electrical pulses are also susceptible to interference from two sources:

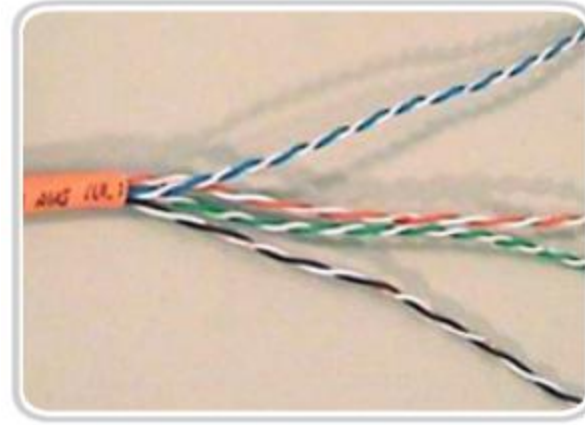
- **Electromagnetic interference (EMI) or radio frequency interference (RFI)** – Potential sources of EMI and RFI include **radio waves** and **electromagnetic devices** such as fluorescent lights or electric motors as shown in the figure.
- **Crosstalk** - Crosstalk is a disturbance caused by the electric or magnetic fields of a signal on one wire to the signal in an adjacent wire. In telephone circuits, crosstalk can result in hearing part of another voice conversation from an adjacent circuit. Specifically, when electrical current flows through a wire, it creates a small, circular magnetic field around the wire which can be picked up by an adjacent wire.



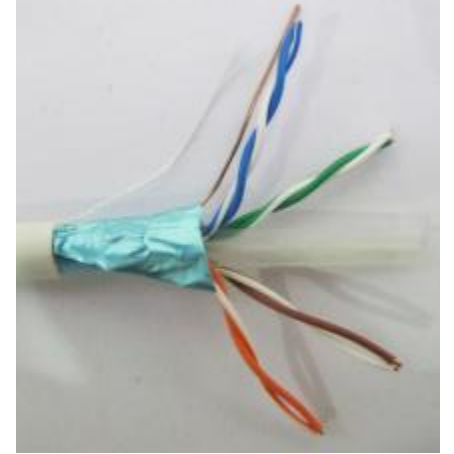
Copper Media

There are four main types of copper media used in networking:

- **Unshielded Twisted-Pair (UTP)**
- **Shielded Twisted Pair (STP)**
- **Foiled Twisted Pair (FTP)**
- **Coax cable**



Unshielded Twisted Pair (UTP) cable



Shielded Twisted Pair (STP) cable



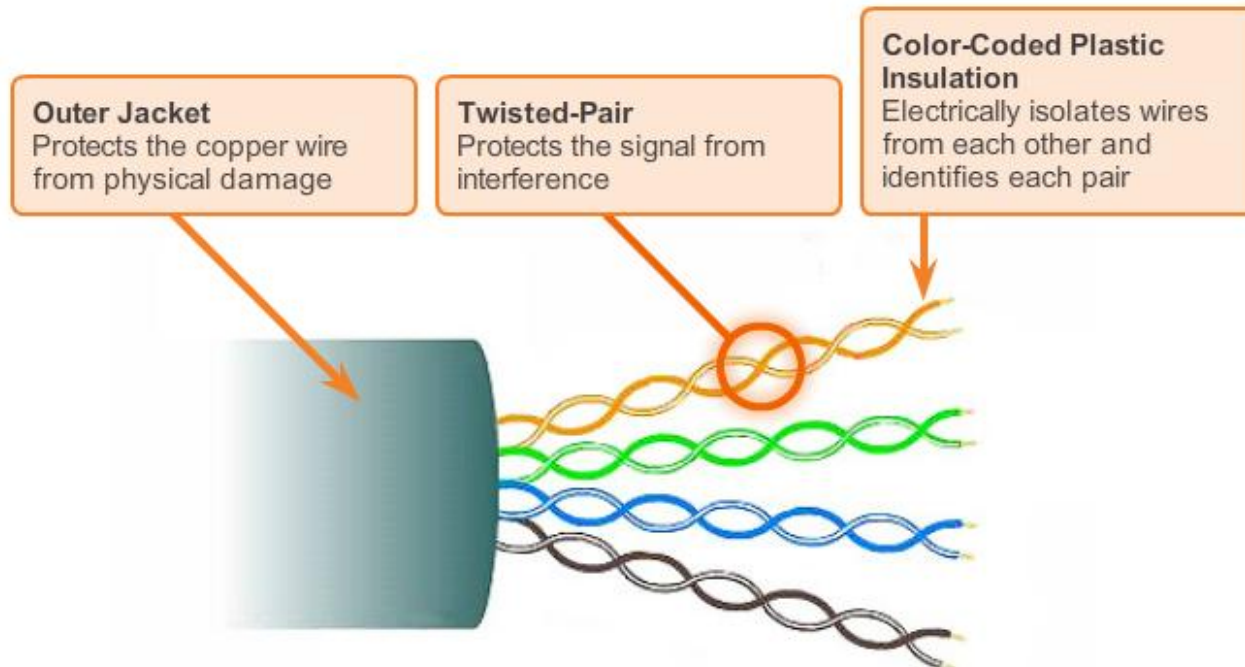
Foiled Twisted Pair (FTP) cable



Coaxial cable

Unshielded Twisted-Pair (UTP) Cable

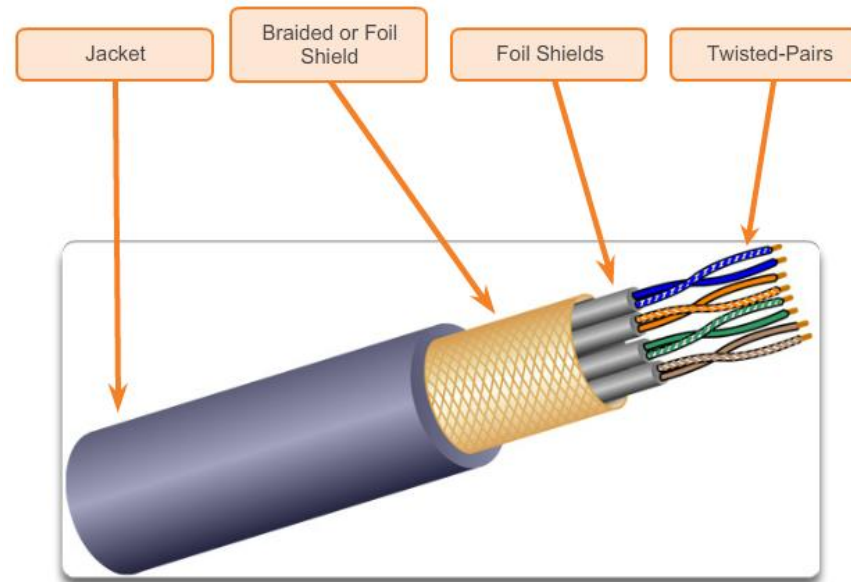
- Unshielded twisted-pair (UTP) cabling is the most common networking media. UTP cabling, terminated with **RJ-45 connectors**



Shielded Twisted-Pair (STP) Cable

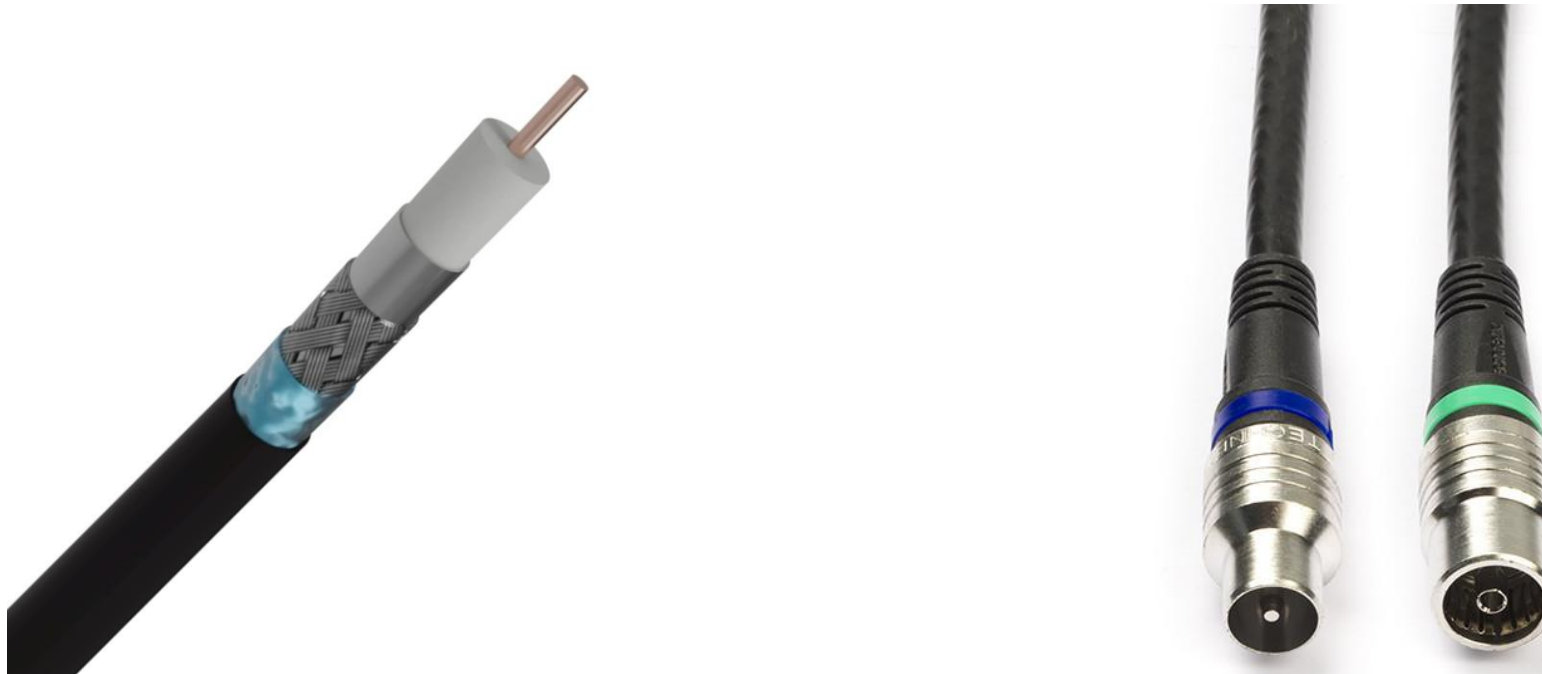
Shielded twisted-pair (STP) and Foiled twisted-pair (FTP) provides **better noise protection** than UTP cabling. However, compared to UTP cable, STP cable is significantly **more expensive** and **difficult to install**. Like UTP cable, STP uses an RJ-45 connector.

Use only for specific situations like high interference environments



Coaxial Cable

Coaxial cable was traditionally used in cable television (75 ohm) capable of transmitting in one direction. It was also used extensively in early Ethernet installations (50 ohm). It is used today by Telenet for the 'last mile' (connection to homes)



Copper Media Safety



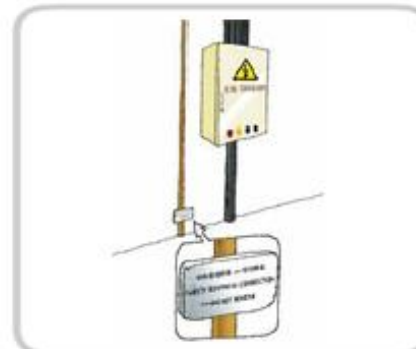
The separation of data and electrical power cabling must comply with safety codes.



Cables must be connected correctly.



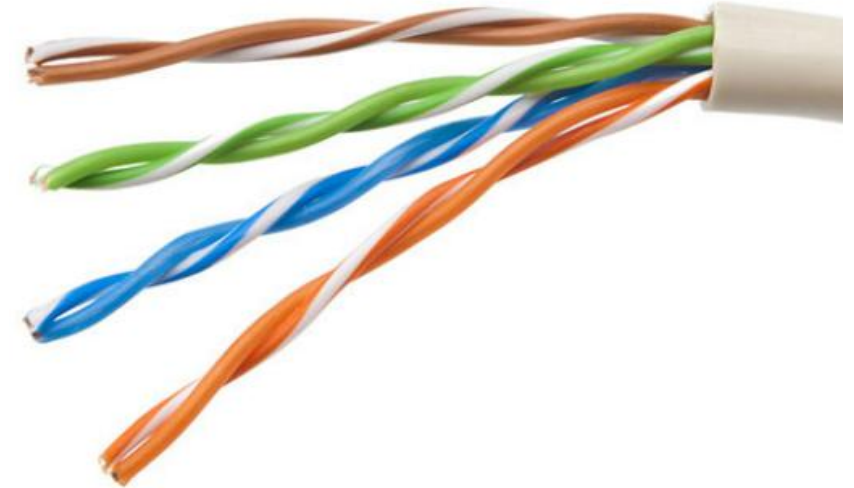
Installations must be inspected for damage.



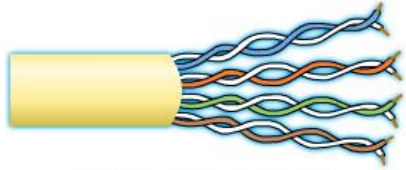
Equipment must be grounded correctly.

Properties of UTP Cabling

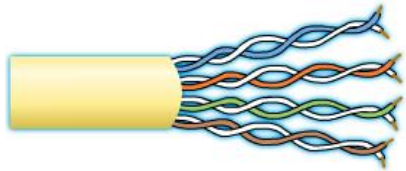
- UTP cable does not use shielding to counter the effects of EMI and RFI. Instead, cable designers have discovered that they can limit the negative effect of crosstalk by:
- **Cancellation:** Designers now pair wires in a circuit. When two wires in an electrical circuit are placed close together, their magnetic fields are the exact opposite of each other. Therefore, the two magnetic fields cancel each other.
- **Varying the number of twists per wire pair:** Notice in the figure that the orange/orange white pairs are twisted less than the blue/white blue pair.



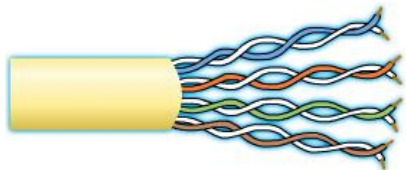
UTP Cabling Standards



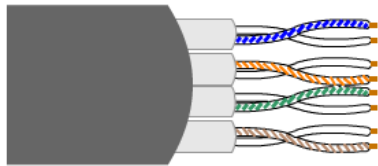
Category 3 Cable (UTP)



Category 5 and 5e Cable (UTP)



Category 6 Cable (UTP)



Category 7 Cable
(ScTP)

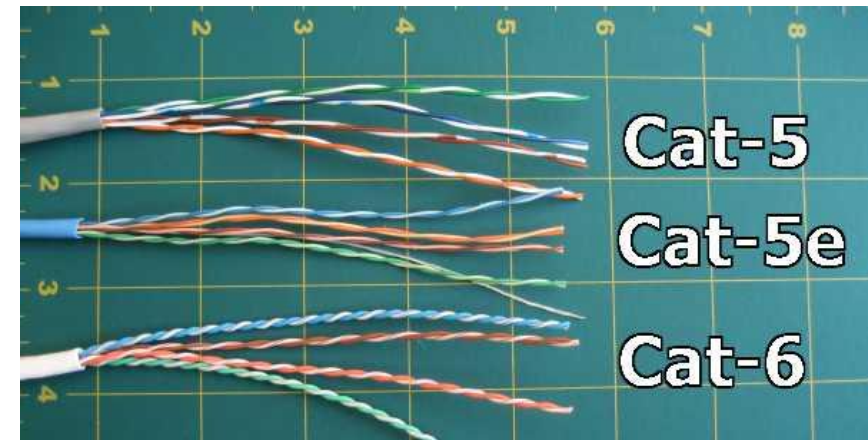
Used for voice communication
Most often used for phone lines

Cat5 supports 100 Mb/s and can support 1000 Mb/s, but it is not recommended
Cat5e supports 1000 Mb/s

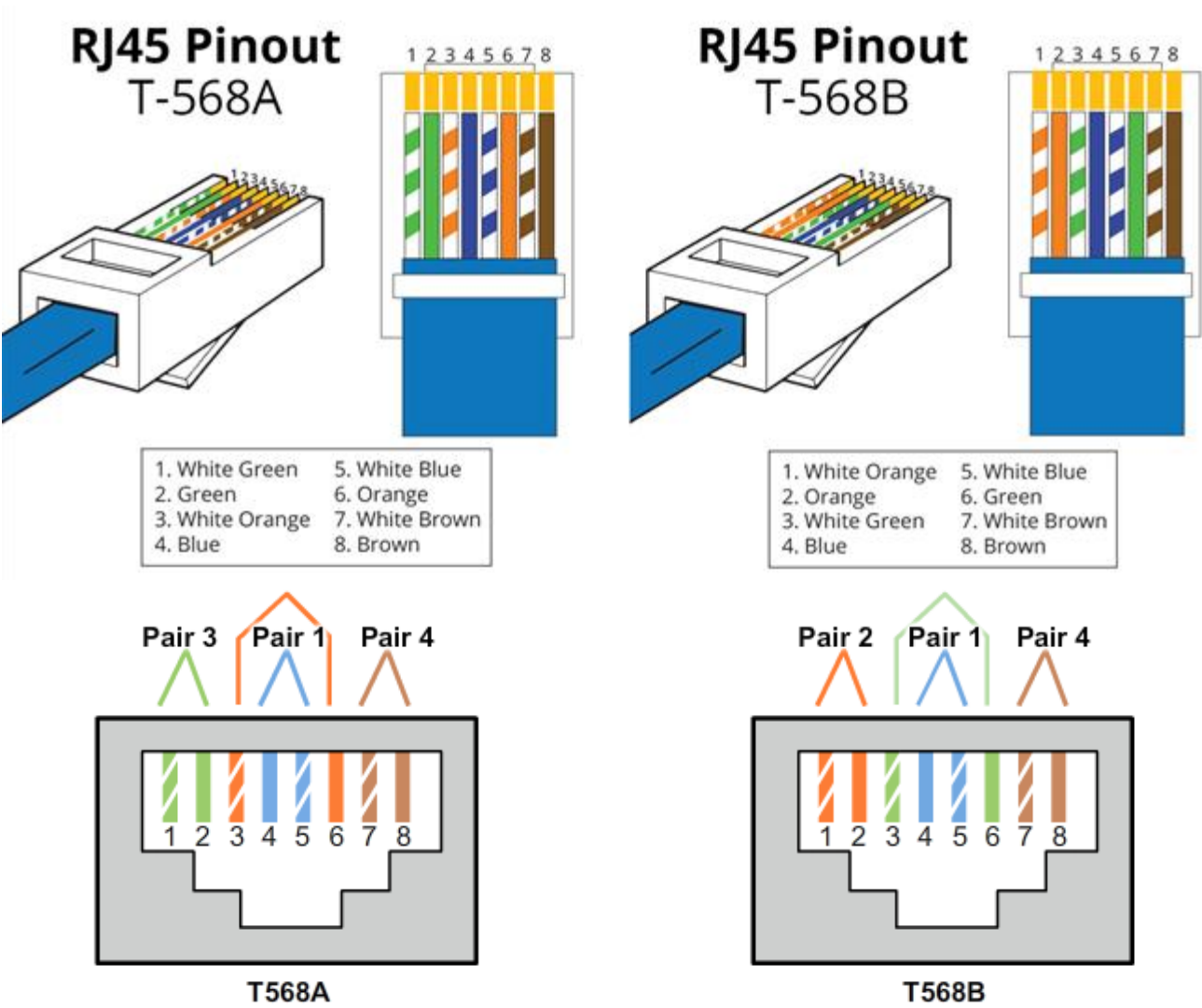
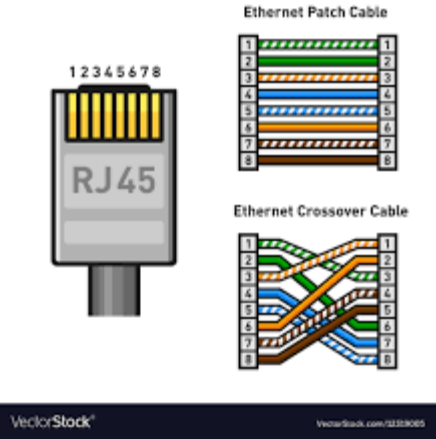
An added separator is between each pair of wires allowing it to function at higher speeds

Supports 1000 Mb/s - 10 Gb/s,
10 Gb/s is not recommended

Supports 10 Gb/s

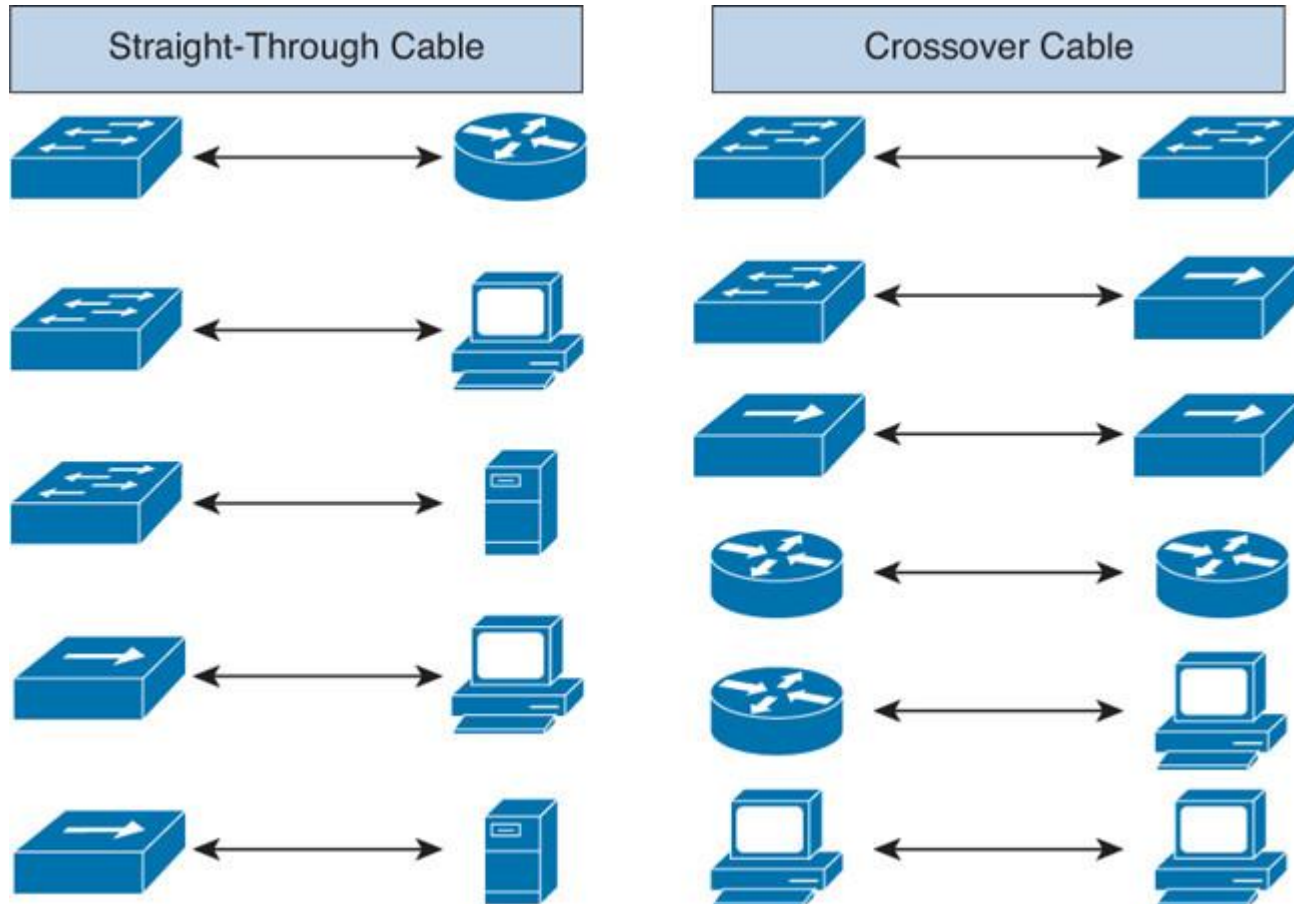


Types of UTP Cable



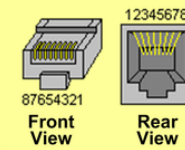
Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or both ends T568B	Connects a network host to a network device such as a switch or hub.
Ethernet Crossover	One end T568A, other end T568B	- Connects two network hosts - Connects two network intermediary devices (switch to switch, or router to router)

Cable usage



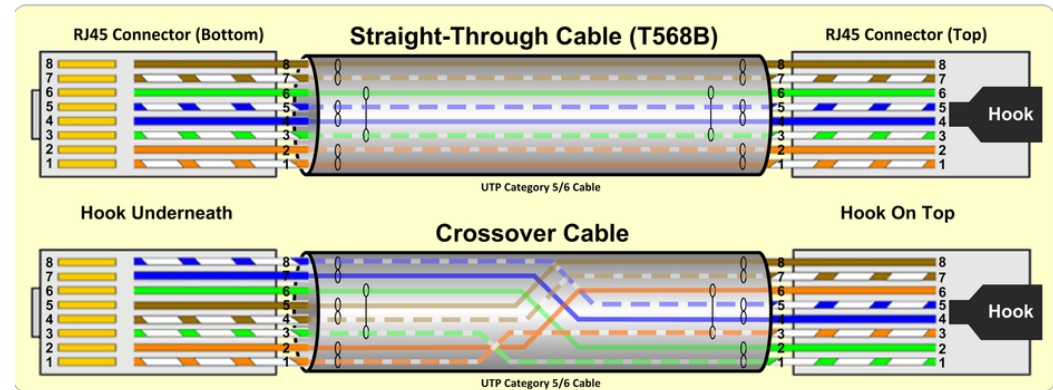
Note: GigaBit Ethernet Requires All 4 Pairs.

RJ45 3D View



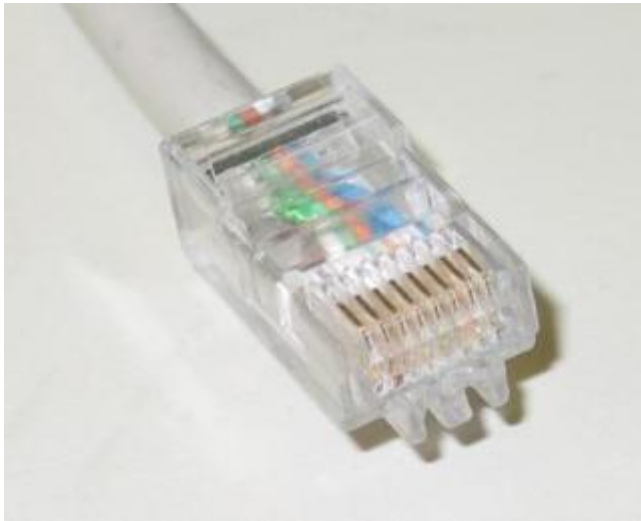
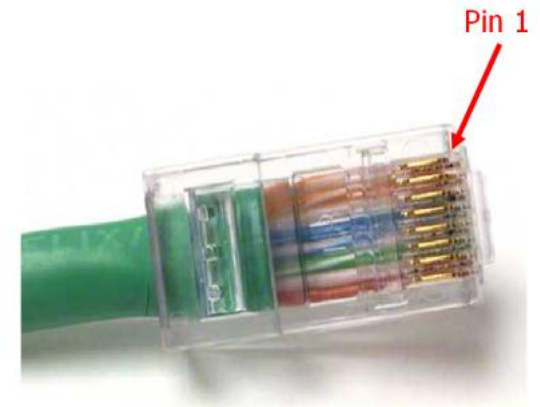
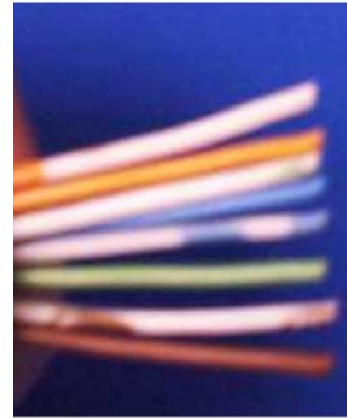
RJ45 - Pinout, Wire Pair Color Coding, and Signal Identification

Pin	T568A	T568B	Signal 10/100BaseTx	Signal 1000BaseT
1	Wht/Grn	Wht/Org	Tx+	TP1+
2	Grn	Org	Tx-	TP1-
3	Wht/Org	Wht/Grn	Rx+	TP2+
4	Blu	Blu	Unused	TP3-
5	Wht/Blu	Wht/Blu	Unused	TP3+
6	Org	Grn	Rx-	TP2-
7	Wht/Brn	Wht/Brn	Unused	TP4+
8	Brn	Brn	Unused	TP4-

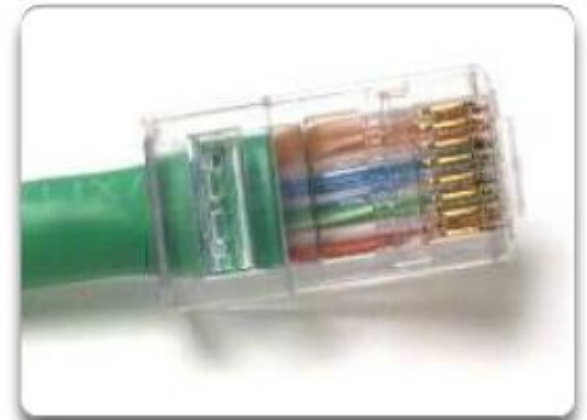


Pin	Name	Description	Cable Color
1	TX_D1+	Tranceive Data+	white/green
2	TX_D1-	Tranceive Data-	green
3	RX_D2+	Receive Data+	white/orange
4	BI_D3+	Bi-directional Data+	blue
5	BI_D3-	Bi-directional Data-	white/blue
6	RX_D2-	Receive Data-	orange
7	BI_D4+	Bi-directional Data+	white/brown
8	BI_D4-	Bi-directional Data-	brown

UTP Connectors



Bad connector - Wires are exposed, untwisted, and not entirely covered by the sheath.



Good connector - Wires are untwisted to the extent necessary to attach the connector.

RJ-45 plugs

RJ-45 UTP Plugs

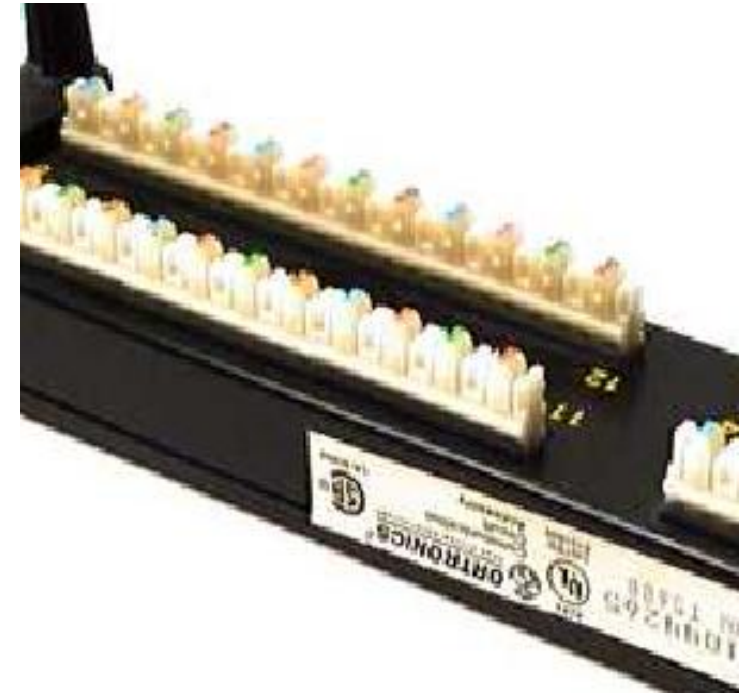


RJ-45 UTP Socket



patch panels

- Patch panels are used to group a work area's connections.
- One side has a place for each RJ-45 plug.
- The back side has punch down blocks that provide connectivity.
- Rack mounted.



Testing UTP Cables

After installation, a UTP cable tester should be used to test for the following parameters:

- Wire map
- Cable length
- Signal loss due to attenuation
- Crosstalk

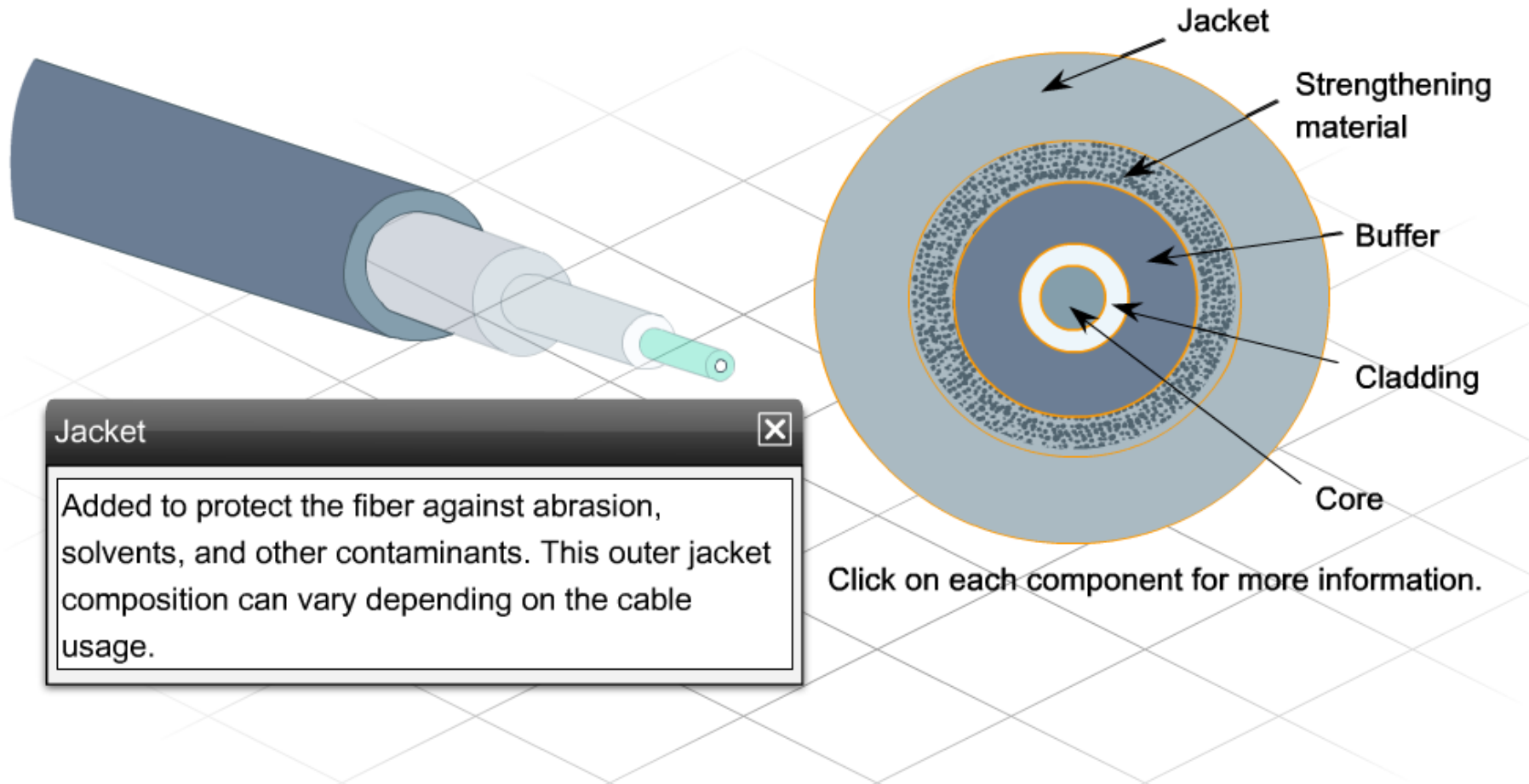


Properties of Fiber Optic Cabling

- Optical fiber is a flexible but extremely thin transparent strand of very pure glass (silica) not much bigger than a human hair.
- Bits are encoded on the fiber as light impulses. The fiber-optic cable acts as a waveguide, or “light pipe,” to transmit light between the two ends with minimal loss of signal.

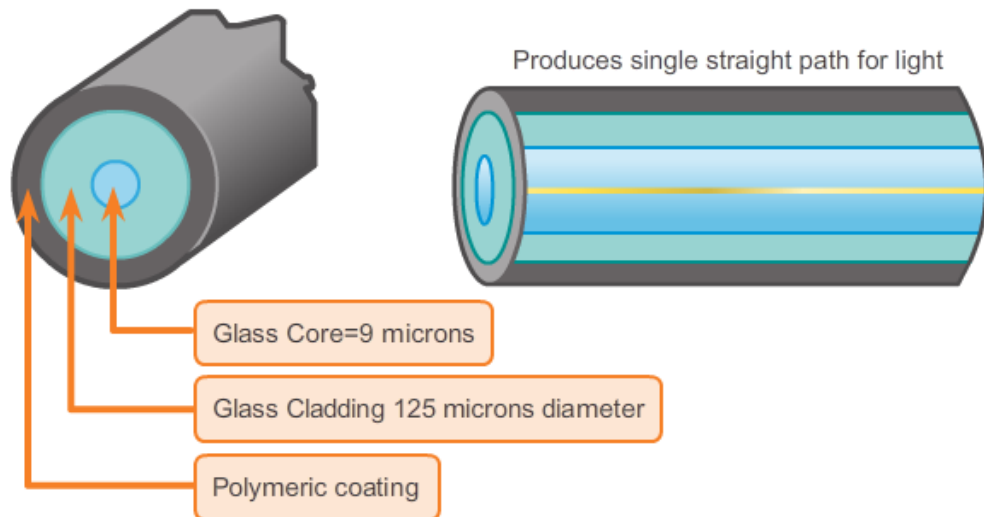


Fiber Media Cable Design



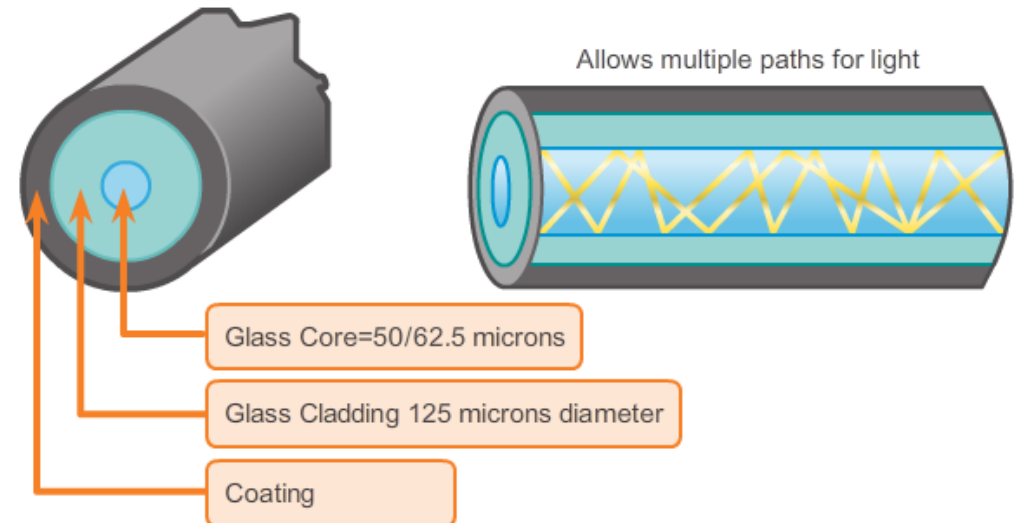
Types of Fiber Media

Single Mode



- Small core
- Less dispersion
- Suited for long distance applications
- Uses lasers as the light source
- Commonly used with campus backbones for distances of several thousand meters

Multimode



- Larger core than single mode cable
- Allows greater dispersion and therefore, loss of signal
- Suited for long distance applications, but shorter than single mode
- Uses LEDs as the light source
- Commonly used with LANs or distances of a couple hundred meters within a campus network

Network Fiber Connectors and patch cords

Common Fiber Patch Cords



SC-SC Multimode Patch Cord



LC-LC Single-mode Patch Cord



ST Connectors



SC Connectors



ST-LC Multimode Patch Cord



SC-ST Single-mode Patch Cord



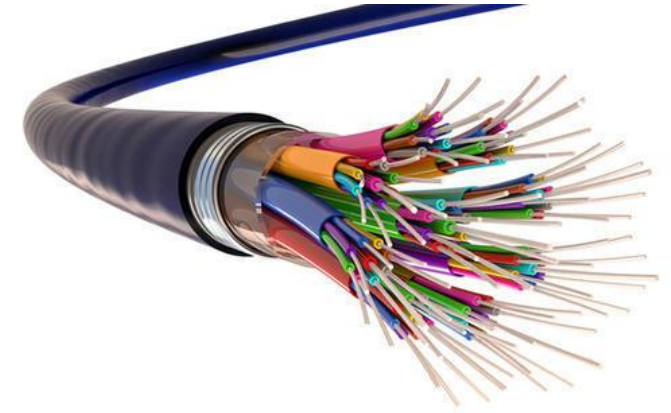
LC Connector



Duplex Multimode LC Connectors

Why is fiber “better”?

Speed through fiber: 70% speed of light
Speed through copper: 50 - 99% speed of light

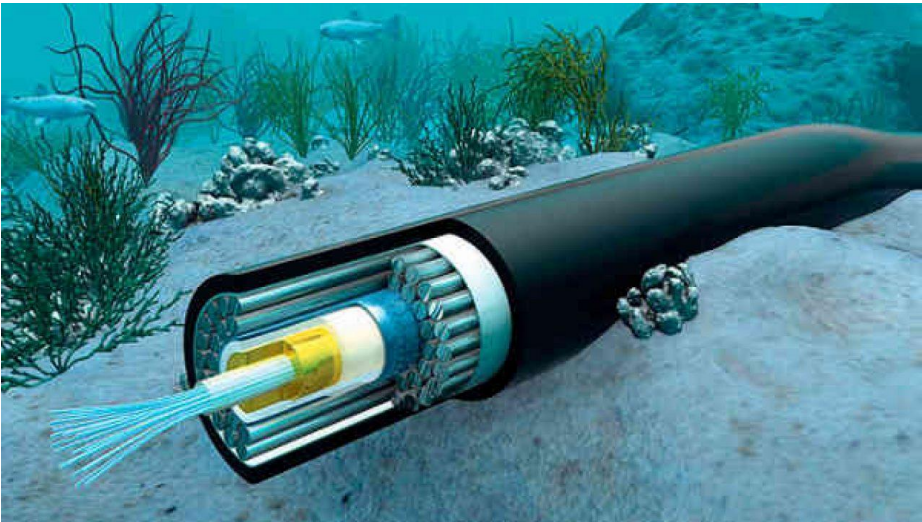
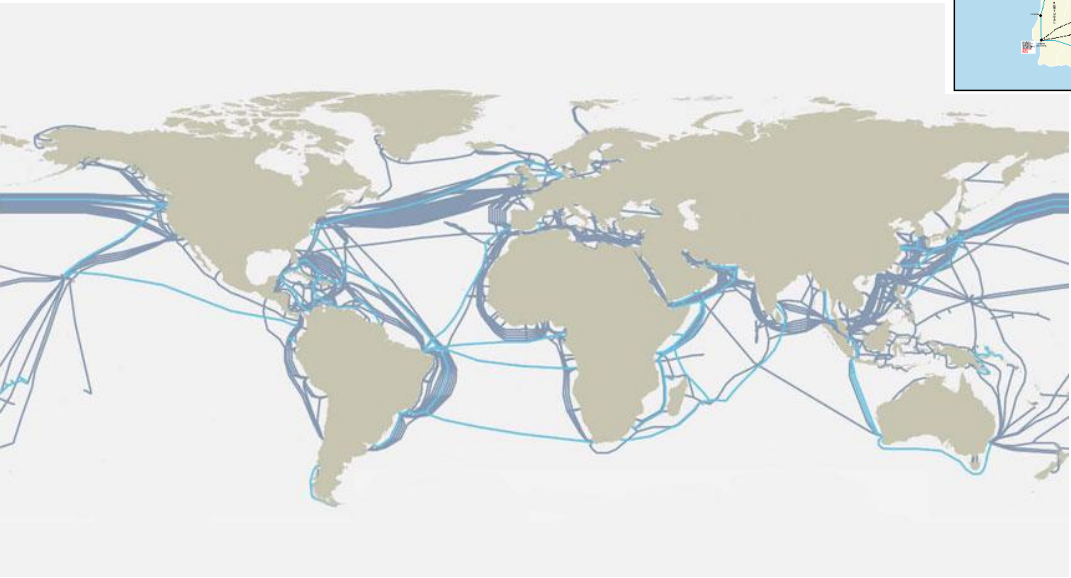
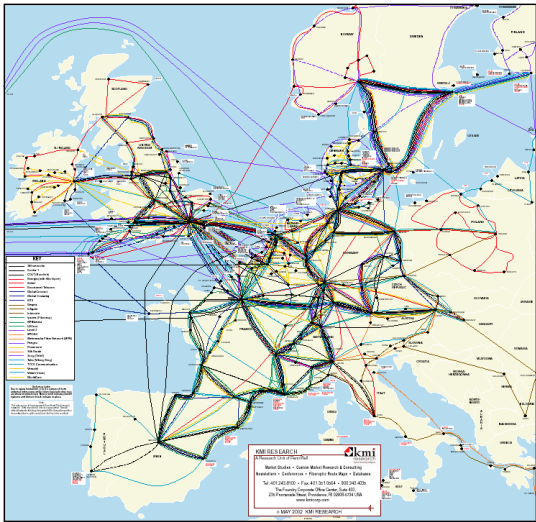


- Immune to electromagnetic interference
- Less loss over distance
- Send at much higher switching speeds
- Fiber cables can be bundled
- Much more modulation channels
- But at higher installation cost per distance
 - What is the required bandwidth?
 - What is the required distance?

- lower data loss
- much longer cables
- higher frequency
- higher bandwidth
- higher bandwidth

What is the cost per distance per bandwidth?

Fiber internet backbone



Hibernia Express	September 2015	4,600 km			Halifax, CA-NS; Cork, IE-M; Brean, GB-ENG	GTT Communications, Inc. (originally Hibernia Networks)
AEConnect (AEC-1)	January 2016	5,522 km	4 × 10 Tbit/s (four strand 100 × 100 Gbit/s)	54 ms	Shirley, US-NY; Killala, IE-C	Aqua Comms®
MAREA	February 2018	6,600 km	160 Tbit/s		Virginia Beach, US-VA; Bilbao, ES-PV	Facebook (25 %), Microsoft (25 %), Telefónica (50 %)
Midgardsormen	Q2 2019 (planned)	7,848 km			Virginia Beach, US-VA; Blaabjerg, DK; Mo i Rana, NO	Midgardsormen®
Dunant	September 2020 (live)	6,400km	250 Tbit/s		Virginia Beach, US-VA; Saint-Hilaire-de-Riez, FR	Google ^{[11][12]}
Grace Hopper	2022 (planned)	6,000km	352 Tbit/s		New York, US; Bude, UK; Bilbao, Spain	Google ^{[13][14]}
Amitié	2022 (planned)	6,600km	320 Tbit/s		Lynn, Massachusetts, US; Bude, UK; Le Porge, France	A consortium comprising Facebook, Microsoft, Aqua Comms, Vodafone (thru Orange) ^[15]

Optic fiber welding

A special tool/machine is needed to weld 2 fiber cables together or to attach a connector



Testing Fiber Cables

Misalignment: The fiber-optic media are not precisely aligned to one another when joined.

End gap: The media does not completely touch at the splice or connection.

End finish: The media ends are not well polished or dirt is present at the termination.



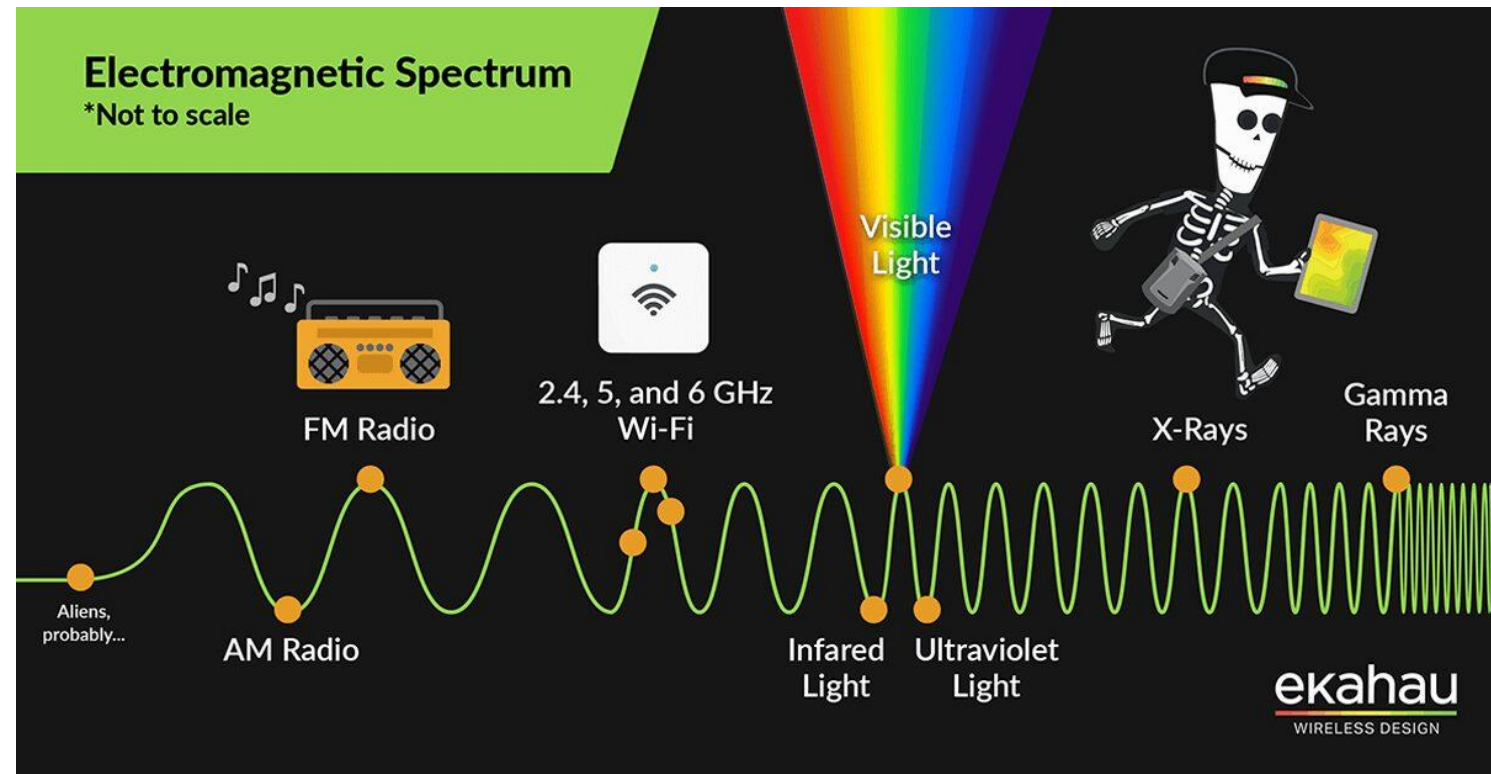
Optical Time Domain Reflectometer (OTDR)

Fiber versus Copper

Implementation issues	Copper media	Fibre-optic
Bandwidth supported	10 Mbps – 10 Gbps	10 Mbps – 400 Gbps
Distance	Relatively short (1 – 100 meters)	Relatively High (1 – 100,000 meters)
Immunity to EMI and RFI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest

Wireless Media

- Wireless media carry electromagnetic signals that represent the binary digits of data communications using radio or microwave frequencies.



Properties of Wireless Media

- **Coverage area:** Wireless data communication technologies work well in open environments. However, certain construction materials used in buildings and structures, and the local terrain, will limit the effective coverage.
- **Interference:** Wireless is susceptible to interference and can be disrupted by such common devices as household cordless phones, some types of fluorescent lights, microwave ovens, and other wireless communications.
- **Security:** Wireless communication coverage requires no access to a physical strand of media. Therefore, devices and users, not authorized for access to the network, can gain access to the transmission. Network security is a major component of wireless network administration.
- **Shared medium:** WLANs operate in half-duplex, which means only one device can send or receive at a time. The wireless medium is shared amongst all wireless users. The more users needing to access the WLAN simultaneously, results in less bandwidth for each user. Half-duplex is discussed later in this chapter.

Wireless LAN

A wireless LAN requires the following network devices:

Wireless Access Point (AP): Concentrates the wireless signals from users and connects, usually through a copper cable. Wireless routers integrate the functions of a router, switch, and access point into one device as shown in the figure.

Wireless NIC adapters: Provides wireless communication capability to each network host.

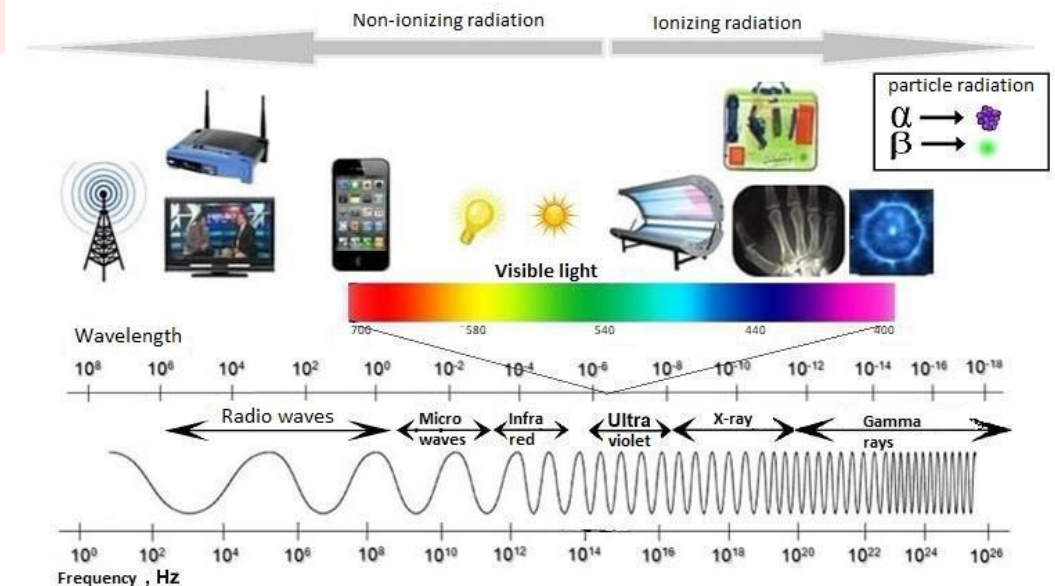


EM wireless bands

Type	Frequencies	Licence
FM Radio	87,5 – 108 MHz	National auctions
3G/4G/5G	800, 1800, 2600 MHz	National auctions
CB Radio	27 MHz	Specific use (Trucks, Hunting, Ships, ...)
Rf electronics/ walkie talkies	433 MHz, 446 MHz	Free to use
WIFI	2,4 and 5 GHz	Free to use
...		

IEEE Standard	WiFi Gen	Year	Frequency	Max PHY Data Rate	Max Range
802.11	-	1997	2.4 GHz	2 Mbps	20m (indoor) 100m (outdoor)
802.11 a	-	1999	5 GHz	54 Mbps	35m (indoor) 120m (outdoor)
802.11 b	-	1999	2.4 GHz	11 Mbps	35m (indoor) 140m (outdoor)
802.11 g	-	2003	2.4 GHz	54 Mbps	38m (indoor) 140m (outdoor)
802.11 n	WIFI 4	2009	2.4/5 GHz	600 Mbps	70m (indoor) 250m (outdoor)
802.11 ac	WIFI 5	2013	5 GHz	6.9 Gbps	35m (indoor)
802.11 ad	-	2012	60 GHz	8.1 Mbps	3.3m (indoor)
802.11 ah	-	2017	Sub 1 GHz	347 Mbps	1km
802.11 ax	WIFI 6	2021	2.4/5/6 GHz	9.6 Gbps	30m (indoor) 120m (outdoor)
802.11 ay	-	2021	60 GHz	303 Gbps	10m (indoor) 100m (outdoor)
802.11 be	WIFI 7	2024	2.4/5/6 GHz	46.1 Gbps	30m (indoor) 120m (outdoor)

The electromagnetic spectrum



802.11 Wi-Fi Standards

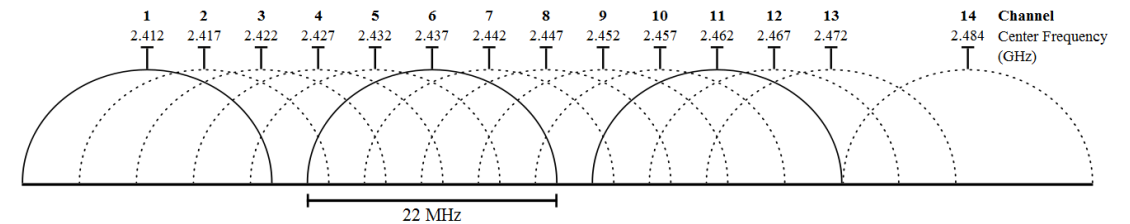
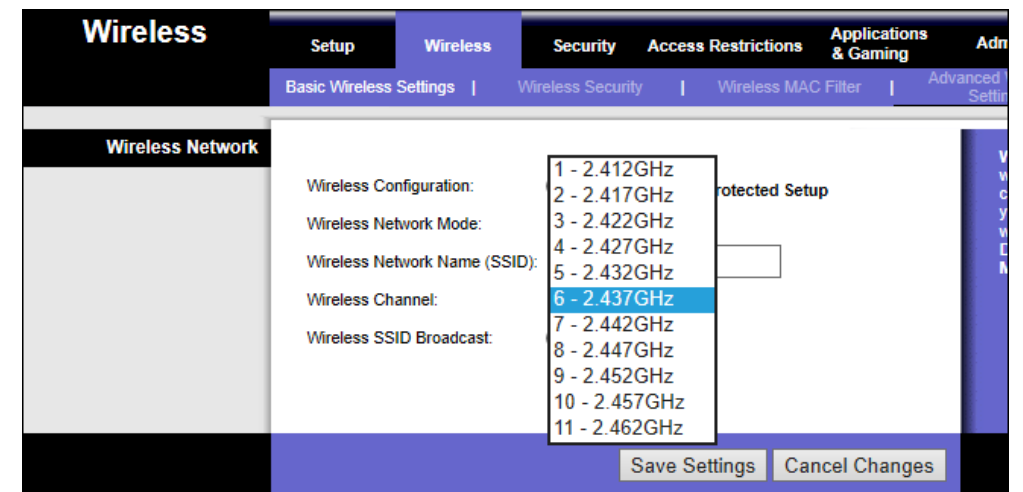
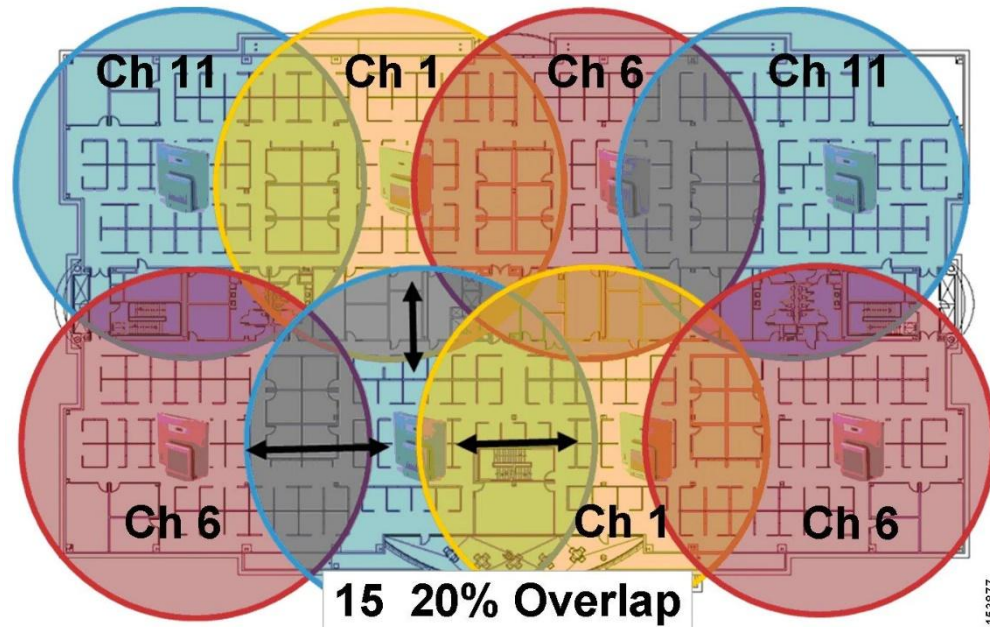
Standard	Maximum Speed	Frequency	Backwards compatible
802.11a	54 Mbps	5 GHz	No
802.11b	11 Mbps	2.4 GHz	No
802.11g	54 Mbps	2.4 GHz	802.11b
802.11n	600 Mbps	2.4 GHz or 5 GHz	802.11b/g
802.11ac	1.3 Gbps (1300 Mbps)	2.4 GHz and 5.5 GHz	802.11b/g/n
802.11ad	7 Gbps (7000 Mbps)	2.4 GHz, 5 GHz and 60 GHz	802.11b/g/n/ac

Wi-Fi characteristics

- Bits are modulated on carrier waves and send over shared medium
 - Different modulation techniques
- Within 2.4GHz and 5GHz, multiple carrier wave frequencies can be used
- Access methods on shared medium
 - Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
 1. Listen on the shared medium if there is someone else sending
 2. Request to send data over medium
 3. Send data
- Shared medium divides time/bandwidth over connected users

Wifi channels

2.4 GHz: 11 channels (partly overlapping)
= 3 usable channels (1,6,11)
5 GHz: 25 usable channels



2.4 GHz (802.11b/g/n)



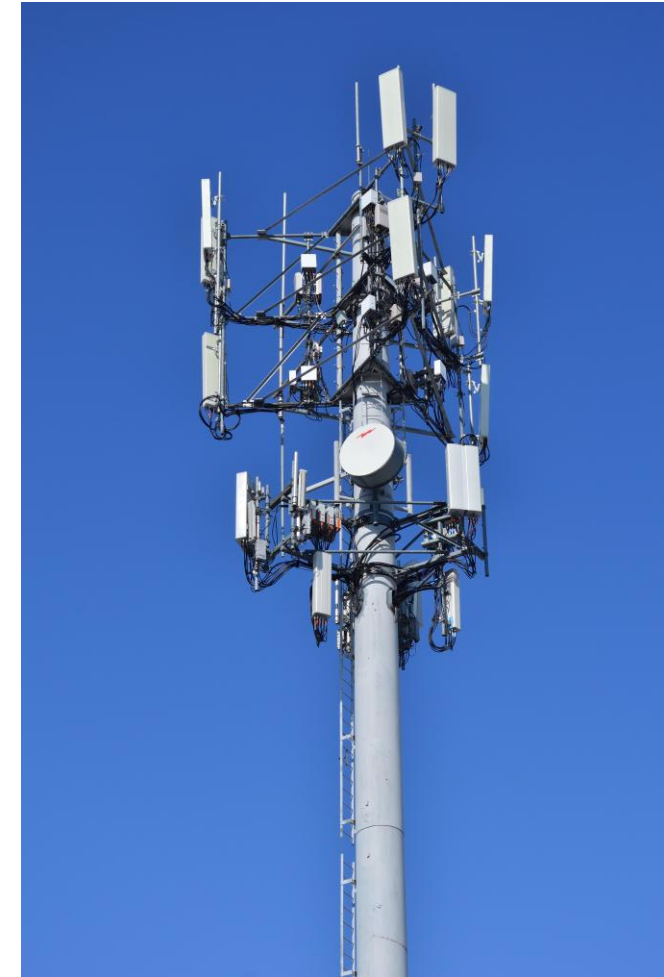
5 GHz (802.11a/n/ac)



Cellular = WAN/mobile carrier technology

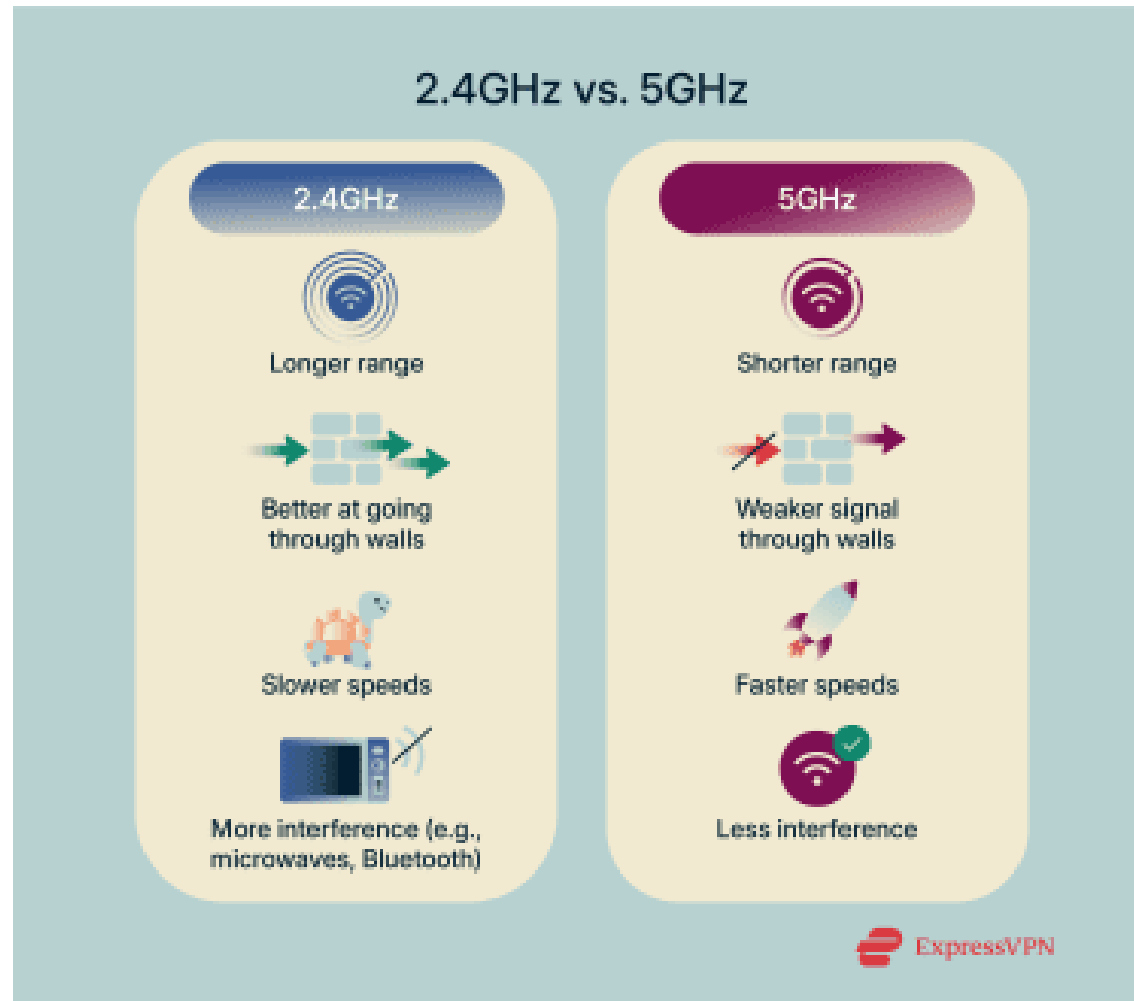
3g/4g/5g described in the 3GPP standard

Feature	Wi-Fi	Cellular
Range	10–50 m	Kilometers
Ownership	Private	Carrier-operated
Frequency	Unlicensed	Licensed
Mobility	Limited	Full mobility
Power	Low	Higher (base stations)



Comparison	2G	3G	4G	5G
Introduced in year	1993	2001	2009	2018
Technology	GSM	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	TDMA, CDMA	CDMA	CDMA	OFDM, BDMA
Switching type	Circuit switching for voice and packet switching for data	Packet switching except for air interference	Packet switching	Packet switching
Internet service	Narrowband	Broadband	Ultra broadband	Wireless World Wide Web
Bandwidth	25 MHz	25 MHz	100 MHz	30 GHz to 300 GHz
Advantage	Multimedia features (SMS, MMS), internet access and SIM introduced	High security, international roaming	Speed, high speed handoffs, global mobility	Extremely high speeds, low latency
Applications	Voice calls, short messages	Video conferencing, mobile TV, GPS	High speed applications, mobile TV, wearable devices	High resolution video streaming, remote control of vehicles, robots, and medical procedures

2.4GHz vs 5GHz



Physical Layer Fundamental Principles

Media	Physical Components	Frame Encoding Technique	Signalling Method
Copper cable	• UTP/FTP/STP • Coaxial • Connectors • NICs • Ports • Interfaces	• Manchester Encoding • Non-Return to Zero (NRZ) techniques • 4B/5B codes are used with Multi-Level Transition Level 3 (MLT-3) signaling • 8B/10B • PAM5	• Changes in the electromagnetic field • Intensity of the electromagnetic field • Phase of the electromagnetic wave
Fiber Optic cable	• Single-mode Fiber • Multimode Fiber • Connectors • NICs • Interfaces • Lasers and LEDs • Photoreceptors	• Pulses of light • Wavelength multiplexing using different colors	• A pulse equals 1. • No pulse is 0.
Wireless media	• Access Points • NICs • Radio • Antennae	• DSSS (direct-sequence spread-spectrum) • OFDM (orthogonal frequency division multiplexing)	• Radio waves